MY CHIREMBA

By

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H200306T

HIT400 Capstone project Submitted in Partial Fulfillment of the

Requirements of the degree of

Bachelor of Technology

In

Software Engineering

In the

School of Information Sciences and Technology

Harare Institute of Technology

Zimbabwe

Supervisor: Ms L Amos

……

May 2024

ABSTRACT

This document proposes the next generation technology that everyone has been waiting for, a novel system that provides medical diagnosis assistance to users. The system is called My Chiremba. The system is designed to help users understand their symptoms and possible causes by guiding them through a series of questions to help narrow down symptoms to a possible diagnosis. The document also entails the root problems that drove the innovation of the system, investigations undergone and the objectives and scope that My Chiremba will have to meet.

PREFACE

In a world where health is paramount, having a trusted ally by your side can make all the difference. Introducing My Chiremba - a revolutionary system designed to transform the way we approach disease diagnosis and daily wellness.

Imagine a tool that listens, understands, and provides precise health insights just by having a conversation. My Chiremba does exactly that. By engaging in a simple dialogue, it gathers your symptoms, analyzes them with precision, and offers you a diagnosis tailored to your unique needs.

But My Chiremba isn't just about diagnosis. It's your daily companion for well-being. From timely health tips to personalized recommendations, it empowers you to take charge of your health journey like never before.

So, whether you're seeking clarity on symptoms or looking for proactive ways to stay healthy, My Chiremba is here to guide you every step of the way.

Welcome to a future where health meets innovation. Welcome to My Chiremba.

ACKNOWLEDGEMENT

First and foremost, I would like to thank the Lord Almighty for giving me the strength and wisdom to be able to complete my capstone project successfully.

I am forever indebted to my parents for showing their unwavering support; be it financial, physical, emotional and spiritual support; during this course of learning curvature.

I would also like to thank my academic supervisor, Ms. L Amos, for giving me guiding and helping me with this project that has come into fruition and also for sharing her expert knowledge which helped in tackling some of the challenges I faced along the way

Justice would not have been served if I forget to thank my school, School of Information Sciences and Technology and also Harare Institute of Technology, for giving me the opportunity to learn a lot under their arm and gain expert knowledge and experience needed in my future endeavors.

The project's success is also a testament to the contributions of the various collaborators, stakeholders and other participants, who helped in some idea generation, data collection and support that they provided.

It is through the collective efforts of all the participants above that drove to the successful completion of the My Chiremba 2024 capstone project.

This is to certify that HIT 400 Project entitled "**MY CHIREMBA**" **has** been completed by **Simon Makowa** (H200306T) for partial fulfilment of the requirements for the award of **Bachelor of Technology** degree in **Software Engineering**. This work is carried out by **him** under my supervision and has not been submitted earlier for the award of any other degree or diploma in any university to the best of my knowledge.

Ms L Amos	**Approved/Not Approved**
Project Supervisor	Project Coordinator
Signature: ……	**Signature: ……**
Date: ……	**Date: ……**

Harare Institute of Technology

success through innovation

Certificate of Declaration

This is to certify that work entitled "MY CHIREMBA "is *submitted in partial fulfillment of the requirements for the award of Bachelor of Technology (Hons) in Software Engineering, Harare Institute of Technology. It is further certified that no part of research has been submitted to any university for the award of any other degree.*

(Supervisor) Signature……..
Date…….

(Mentor) Signature……..
Date……

(Chairman) Signature……..
Date……..

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CHAPTER 1: INTRODUCTION

Background

The rapid pace of technological innovation has transformed nearly every aspect of our lives, revolutionizing the way we work, communicate, and even manage our health and wellbeing. In the field of healthcare, the integration of cutting-edge tools andtechniques has ushered in a new era; of more efficient, effective, and personalized care; that promises to deliver more accurate diagnoses, targeted treatments, and preventive care tailored to the unique needs of each individual.

However, despite these strides, a stark reality persists. In many communities around the world, including Zimbabwe, a significant portion of the population continues to face daunting barriers when it comes to accessing quality healthcare services. Whether it's the prohibitive cost of visiting a doctor or the lack of essential medical equipment and diagnostic tools in underfunded healthcare facilities, these challenges have left far too many Zimbabweans with limited options for managing their health.

Problem Statement

- Many patients lack access to medical professionals who provide accurate and timely diagnosis and treatment. This can be due to financial difficulties or the patients staying in remote or underserved areas where there are no nearby clinics or hospitals. (**It is costly**)
- Healthcare providers often face time constraints in conducting comprehensive assessments and consultations with patients. The limited time available can lead to

delays in treatment and ultimately poorer health outcomes for clients. (**It takes too much time**)

Objectives

- To provide real-time interaction which uses natural language processing for semantic analysis of the description of symptoms
- To provide a functionality for diagnosing a user based on given symptom
- To provide a functionality for giving the users daily health advices

Hypothesis and Justification

The hypothesis for the project "My Chiremba" is that by developing a medical assistant app capable of accurately diagnosing patients' diseases based on symptoms provided, we can improve the efficiency and accessibility of healthcare services, leading to earlier detection and treatment of illnesses which will improve the quality of healthcare in Zimbabwe.

Justification:

1. Improved Healthcare Accessibility: Many people in Zimbabwe do not have access to healthcare facilities or cannot afford medical care. By leveraging the proposed technology, the My Chiremba can help bridge this gap by providing healthcare assistance to individuals who may have limited access to medical professionals or healthcare facilities, particularly in remote or underserved areas.

2. Early Disease Detection: The timely identification and diagnosis of diseases significantly impact patient outcomes. My Chiremba can serve as an efficient screening tool, enabling individuals to receive preliminary assessments and recommendations promptly, leading to early detection and intervention. With this the app can also help reduce the burden on healthcare facilities by providing pre-primary care for quick guidance so that red flags are timely detected by doctors and patients

3. Personalized Healthcare: The app can utilize artificial intelligence algorithms to analyze individual symptoms, medical history, and other relevant factors to provide personalized insights and recommendations to users, empowering them to make informed decisions and take proactive measures for their health.

Proposed Tools

1. Python Language: This Programming language will be used for the development of the system. The system's interface will be built using the Flask package; and the language will be resposnible for all the processes

2. Machine Learning (ML) Algorithms: The KNN algorithmscan be utilized to train the app to recognize patterns and make accurate diagnoses based on symptom inputs and the NLP can be used for semantic analysis during the conversation

3. Secure Data Storage: Xammp will be used for storing user information and also the conversations that have been done on the system by a user

Feasibility Study

1. Economic Feasibility:

My Chiremba can be developed at a relatively low cost, the only hardware required will be the cell phones or laptops which people already have and it can be built using open-source software which is free. The app can be monetized through advertisements, or subscription models. The cost of developing the app can be offset by the potential benefits it provides to the healthcare industry in Zimbabwe therefore the economic feasibility passes.

2. Technical Feasibility:

The development of My Chiremba requires no hardware requirements, software requirements of python (which is open source), and other complimentary software needed for efficient

programming process which is totally free, and the technical requirements are the development skills. The requirements needed can be easily met therefore technical

3. Operational Feasibility:

My Chiremba can be operated with minimal training and can be used by anyone with a smartphone. The app can reduce the time a patient waits to be diagnosed and can improve healthcare accessibility to patients therefore the operational feasibility passes

Project Plan

Gantt Chart

TITLE	AUG	SEPT	OCT	NOV	DEC	JAN	FEB	MAR	APR
Research									
Documentation									
Planning									
Design									
Development									
System Testing									

CHAPTER 2: LITERATURE REVIEW

Introduction

This literature review aims to explore the existing research and applications of artificial intelligence in healthcare, highlighting their impact on improving diagnosis accuracy, enhancing patient engagement, and addressing the challenges faced by healthcare professionals. By critically examining the literature, we can gain insights into the current state of medical assistant technology and how we can apply them in our project, identify research gaps, and unlock opportunities for further advancements in this field.

Related Work

Articles

- An article was published in the National Library of medicine website with the title *Chatbots and Conversational Agents in Mental Health: A review of the Psychiatric Landscape*. The results of the study showed that conversational agents had potential for benefit in psychoeducation and self-adhering and also satisfaction rating of chatbots were extremely high, therefore suggesting that it would be an effective and at the same time enjoyable tool in psychiatric treatment
- *Artificial Intelligence-Based Symptom Checker Systems: A Review and Critical Appraisal* by Johnson et al. (2020): This review article explores the existing AI-based symptom checker systems and evaluates their accuracy, usability, and clinical relevance. It discusses the challenges associated with symptom analysis and provides insights into the potential of AI-driven systems for improving diagnosis and decision-making in healthcare.
- Another article on with the title *Mobile Health Applications for Diagnosis and Management of Medical Conditions* by Lalloo et al. (2019) examines the use of mobile health (mHealth) applications for diagnosis and management of medical conditions. It discusses the role of mHealth apps in symptom tracking, self-diagnosis, and remote monitoring of patients. The review also addresses the challenges and opportunities associated with mHealth app adoption in healthcare.

Systems

- Viz.ai is a mobile app that uses artificial intelligence to detect diseases such as intracerebral hemorrhage and vessel occlusions. This app also provides mobile imaging and HIPPA-complaint messaging to help synchronize workflow across many specialties for diseases such as stroke.
- EpicCare is an electronic record system that allows health professionals to update patient's records in real time and also allows for the integration of various medical devices and data sources.
- Infermedica offers an AI-driven symptom checker that enables users to describe their symptoms. The system analyzes the input and provides potential diagnoses and recommendations for further medical evaluation.
- Buoy Health is a web based ai health assistant that provides personalized clinical support and helps people make the best decisions about their health. It gives people insight on what to do when they are sick

Conclusion

In conclusion, the integration of real-time assistants and AI technologies in healthcare holds significant promise for transforming the way we approach disease diagnosis and patient care. The reviewed literature demonstrates the potential of these technologies in improving diagnosis accuracy, reducing healthcare disparities, and empowering individuals to take an active role in managing their health. However, several challenges remain, including ensuring privacy and data security, addressing ethical considerations, and refining the accuracy and reliability of the symptom analysis. Future research should focus on developing robust medical assistant systems that can handle complex medical queries, incorporate multimodal inputs, and integrate with existing healthcare infrastructure seamlessly. By embracing the potential of medical assistants in healthcare, we can pave the way for a more patient-centered, accessible, and efficient healthcare ecosystem that ultimately improves health outcomes and enhances the overall well-being of individuals.

CHAPTER 3 - ANALYSIS

Information Gathering Tools

Questionnaires

A questionnaire is a document with carefully crafted questions meant for the respondent to answer in the empty spaces provided.

Their advantage is that:

- It is effective when collecting a lot of information
- The questions are made simple to answer
- It is cheap as the researcher does not have to incur too much transport costs
- It saves time as questionnaires are distributed and filled in later

But their disadvantage is that:

- Some questions are left blank
- Some questionnaires are not returned

PATIENT QUESTIONAIRE FORM FOR APPOINTMENT AND TREATMENT PROCEDURE

Name …….

Date …….

The questions below are going to be used to access the current medical treatment process and the evaluate the need for a new automated system which helps with that so may you please be sincere in answering the question. Please answer all of the questions so that we may have a better reflection, thank you.

ON QUESTION 1 TO 5 TICK THE ANSWER THAT YOU WOULD GO FOR

1. How do you typically schedule appointments for medical diagnosis?

   A. Phone call
   B. In-person visit
   C. Online portal
   D. Other (please specify): _____

2. What challenges do you face during the appointment scheduling process? (Select all that apply)

   A. Difficulty reaching the healthcare provider

B. Long waiting times to get an appointment
C. Limited availability of preferred healthcare providers
D. Other (please specify): _____

3. On average, how long do you typically wait for your appointment?

A. Less than 5 minutes
B. 5-30 minutes
C. 30 minutes to 1 hour
D. More than 1 hour

4. How would you rate the overall efficiency of the current procedure for medical diagnosis?

A. Very efficient
B. Somewhat efficient
C. Neutral
D. Inefficient

5. Would you support the introduction of a medical system that assists with the medical diagnosis and treatment process to compliment the manual system?

A. Yes
B. No

ON QUESTION 6 TO 9 YOU ARE REQUIRED TO EXPLAIN

6. What do you like about the manual system?

_____
_____
_____
_____

7. What do you dislike about the current manual system?

_____
_____
_____
_____

8. Is there anything else you would like to share regarding your experience with the current system?

_____
_____
_____
_____

9. What improvements or changes would you suggest to enhance the current procedure for medical diagnosis?

_____
_____
_____
_____

Interviews

An interview is a face-to-face communication between two people to gather information.

The benefits of this method are:

- Information can be collected from even the illiterate (some construction employees that do manual labour)
- The researcher can probe to get more information or get clarification on unclear answers
- Encourages good rapport between the researcher and the respondent
- Non-verbal gestures help the researcher to see if the respondent is not lying

The method also has its disadvantages which are:

- It is time consuming
- Good interview techniques are required as failure may lead to getting biased information

<u>Sample questions on the interview</u>

1. What is your name?
2. What is your age?
3. How frequently do you seek medical assistance?
4. What are the primary reasons for your medical visits?
5. How do you currently schedule appointments with healthcare providers?
6. How do you access your medical records and health information?
7. How do you view the current manual system?
8. What improvements would you want for the current system?

Record inspection

This is scrutinizing system documents so as to gain information.

The pros of the system are:

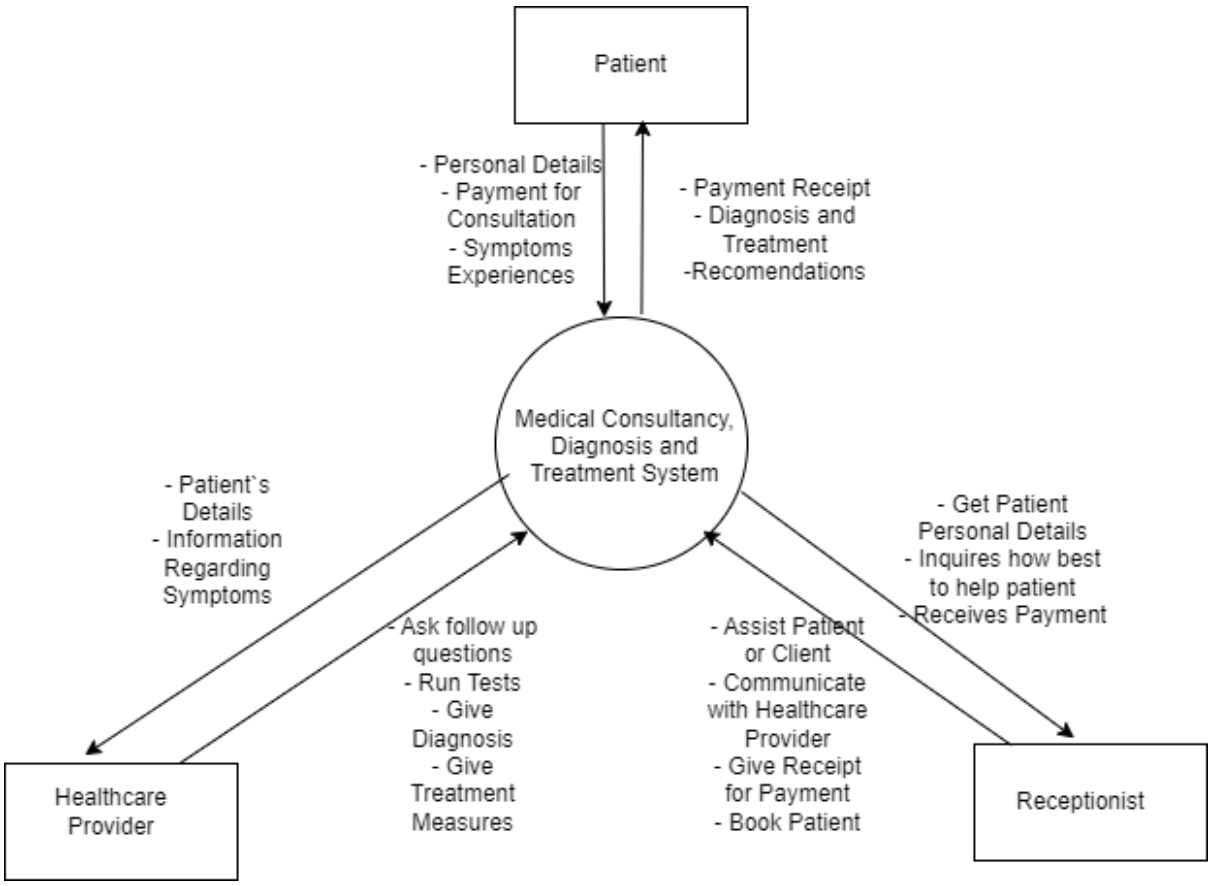
- The researcher gets first-hand information
- Accurate information is collected
- One can leave out information which is not necessary

The cons are:

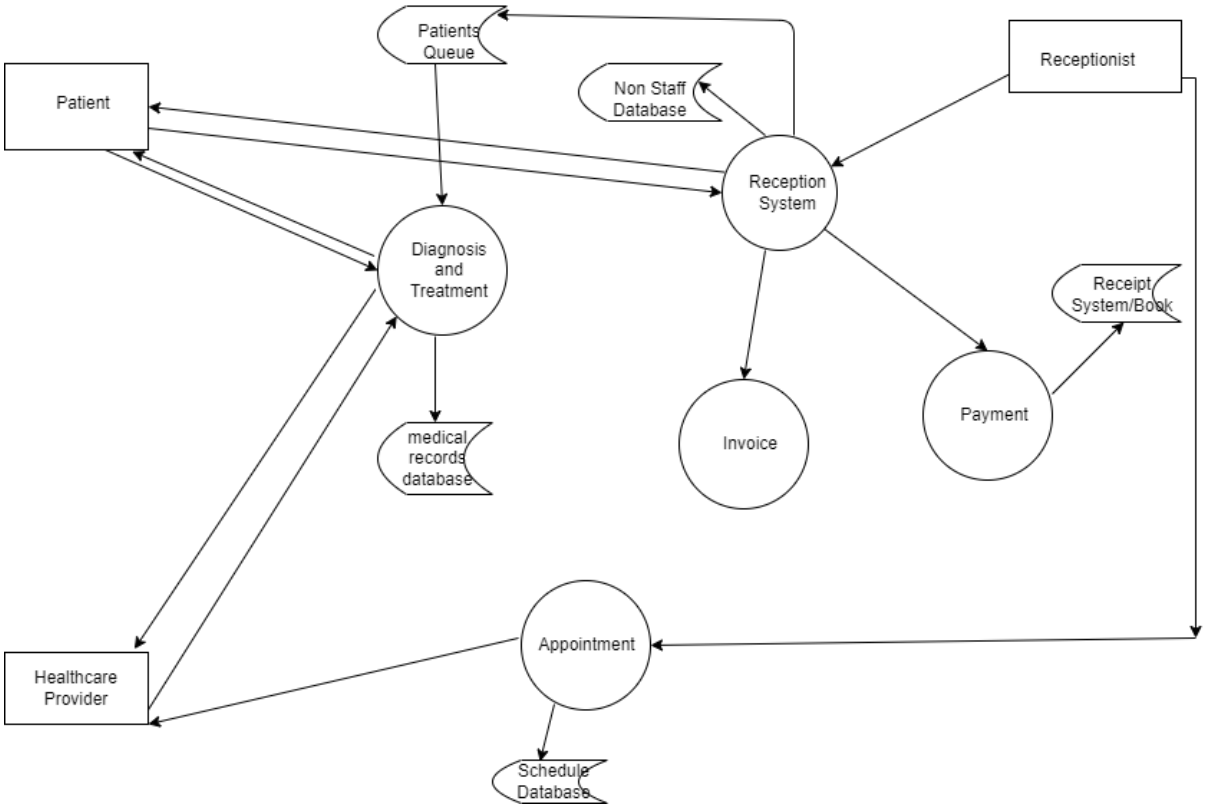
- Confidential files are not given to the researcher

Data analysis

Context Level Data Flow Diagram for the existing system (DFD)



Level 2 Data Flow Diagram for the existing system (DFD)



Evaluation of Alternatives Systems

The problem statements given above can be solved by implementing various techniques and systems and these alternative methods include the ones below:

Improving the Current System

This would involve attempting to lower medical costs by implementing policies that address social determinants of health such as poverty or by negotiating lower prices for medical supplies and services with providers. Another way would involve increases the medical staff in hospitals and clinics therefore reducing the service time for patients in queues.

Challenge: While this solution might be appealing as it might seem the one with least changes but it is difficult to implement because such policies have to go through a lot of scrutiny and approval from higher offices and this will take a lot of time and might also be rejected.

Outsourcing Software Solutions and Introducing Patients to the Solutions

This would involve recommending patients to use some various systems that are found on the internet that include Viz.ai, Infermedica, Bouy Health and EpicCare. These systems provide

functionalities that help doctors to discover patients` illness and provide an electronic healthcare database for the patients.

Challenge: This alternative is beaten by the proposed system because it is too expensive. Most systems need to be purchased for use and the ones that do not require large sums of monthly subscriptions. This will then defeat the issue of trying to lower costs. These systems are also doctor centered therefore they mostly help doctors and not patients

Proposed Solution`s Superiority

My Chiremba stands out as the best solution compared to the alternatives due to its comprehensive set of features and capabilities. It combines advanced real-time communication, natural language processing, and artificial intelligence to provide accurate medical information, diagnostic support, personalized recommendations, and efficient healthcare management.

- It is a personalized solution for patients in Zimbabwe

- The system is centred on the patients therefore it reduces the need for regular appointments with doctors for minor symptoms

- It allows diagnosis in the comfort of the patients` homes, it is a mobile service

- It has a rich set of features that help with medical diagnosis and advices

Functional Analysis of Proposed System

Functional Requirements

1. Real-time Communication and Natural Language Processing:

   - The system should provide real-time feedback to the user during a convesation.

   - It should employ natural language processing techniques to understand user input and provide appropriate responses.

2. Medical Information and Guidance:

   - The system should have a comprehensive medical knowledge database to provide accurate and up-to-date information on symptoms, conditions, treatments, and medications.

- It should offer personalized recommendations based on user-specific information and context.

3. Diagnostic Support and Decision-Making:

- The system should analyze user symptoms, medical history, and risk factors to provide preliminary assessments and potential diagnoses.

- It should assist healthcare providers and patients in making informed decisions about further evaluation and treatment options.

4. Personal Health Monitoring:

- The system should enable users to track and monitor their health conditions, vital signs, and medication adherence.

- It should provide reminders for medication intake and offer lifestyle recommendations based on user-specific health data.

Non-Functional Requirements

1. Usability and User Experience:

- The system should have a user-friendly interface that is easy to navigate and interact with.

- It should provide a seamless and intuitive user experience, ensuring that users can quickly access the desired information or perform tasks.

2. Performance and Scalability:

- The system should respond quickly and provide timely information and assistance to users.

- It should be able to handle multiple user requests simultaneously and scale to accommodate increasing user demand.

3. Security and Privacy:

- The system should ensure the security and privacy of user data, adhering to industry standards and regulations.

- It should employ encryption, access controls, and other security measures to protect sensitive user information.

4. Reliability and Availability:

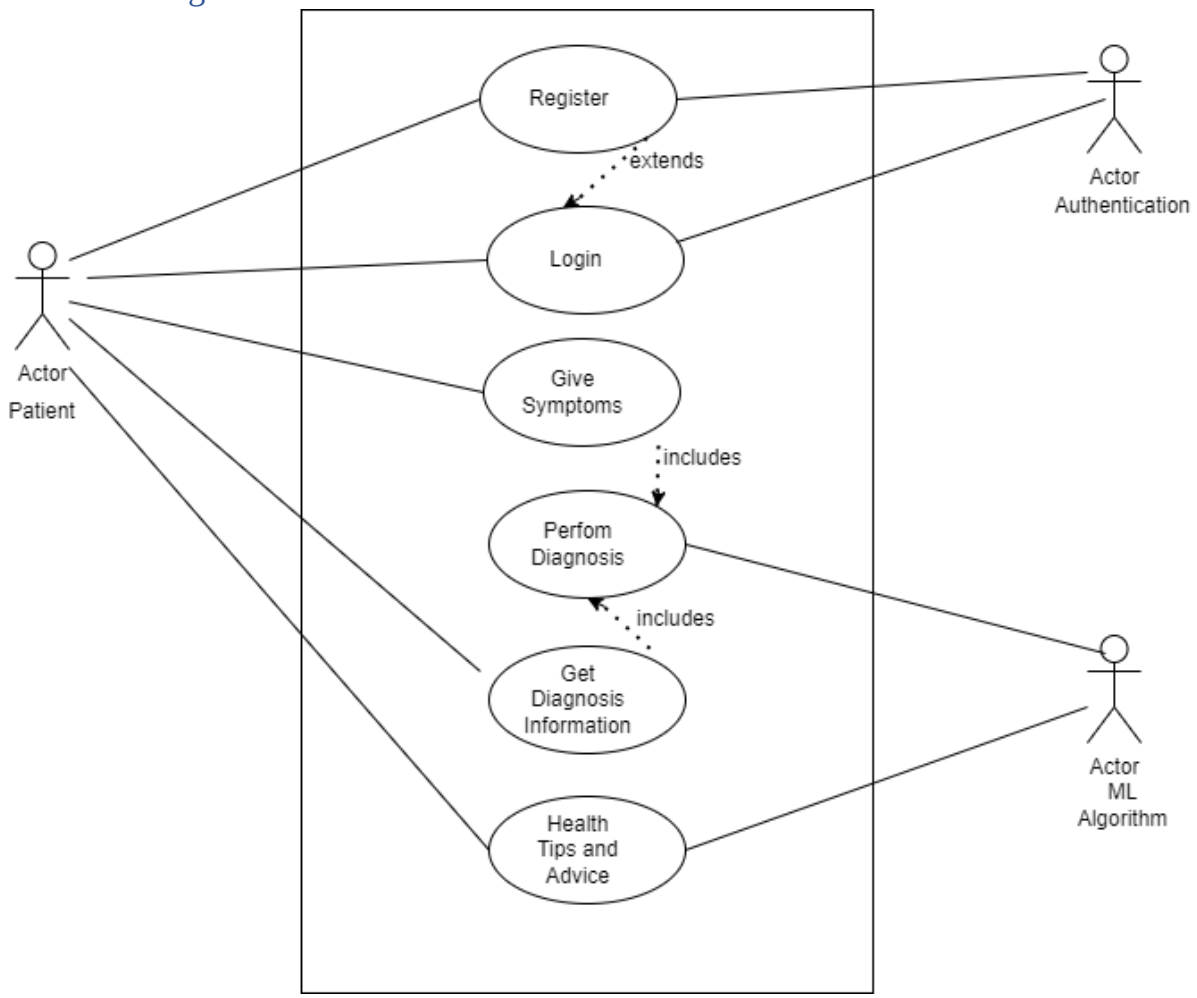
- The system should be reliable, with minimal downtime and system failures.

- It should be available to users 24/7, ensuring uninterrupted access to essential healthcare information and services.

5. Compliance:

- The system should comply with relevant regulatory requirements and healthcare industry standards, ensuring data protection and patient confidentiality.

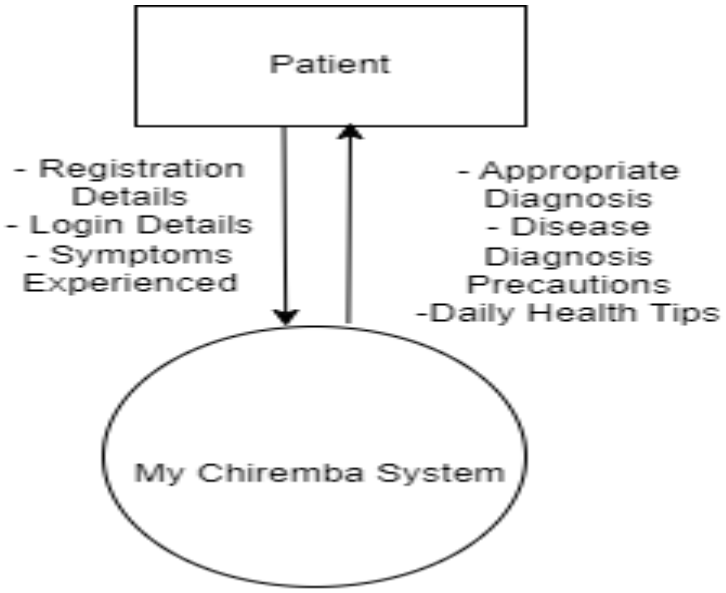
User Case Diagrams



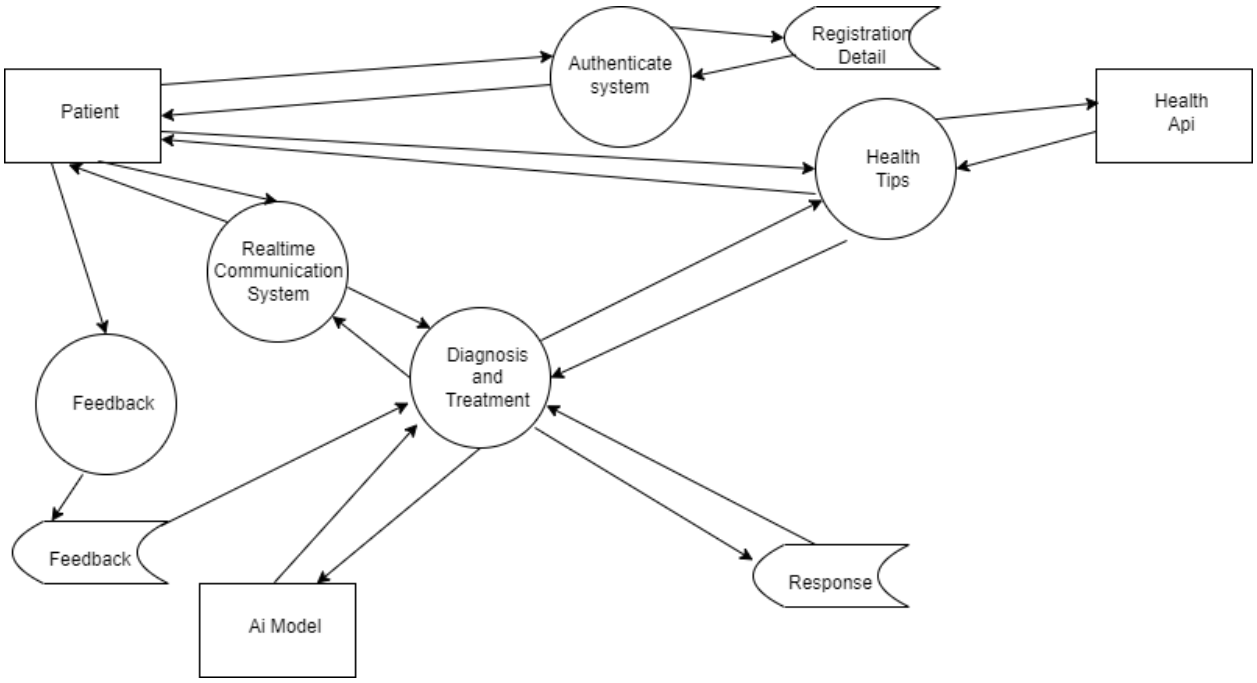
CHAPTER 4: DESIGN

Systems Diagrams –Using UML Context diagrams, DFD, Activity diagrams

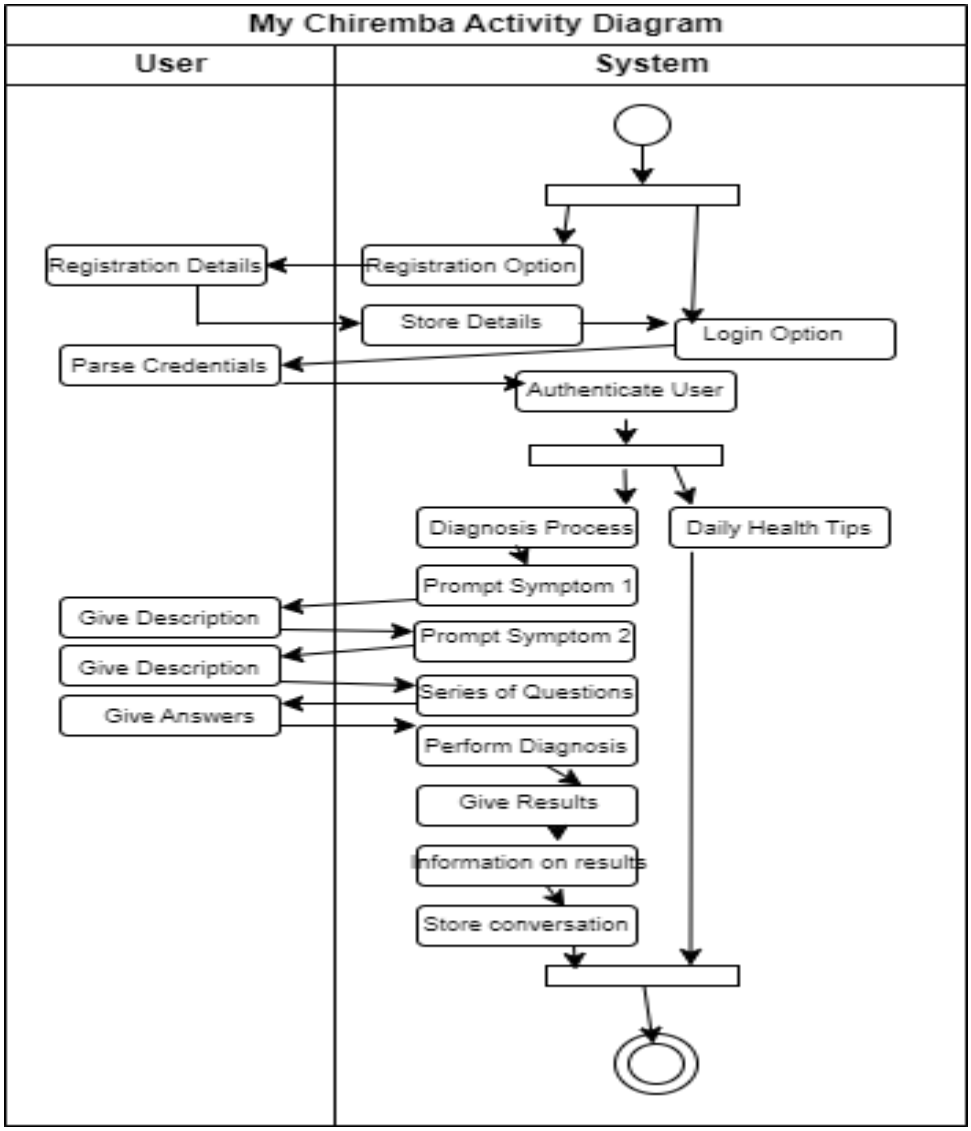
Context Diagram



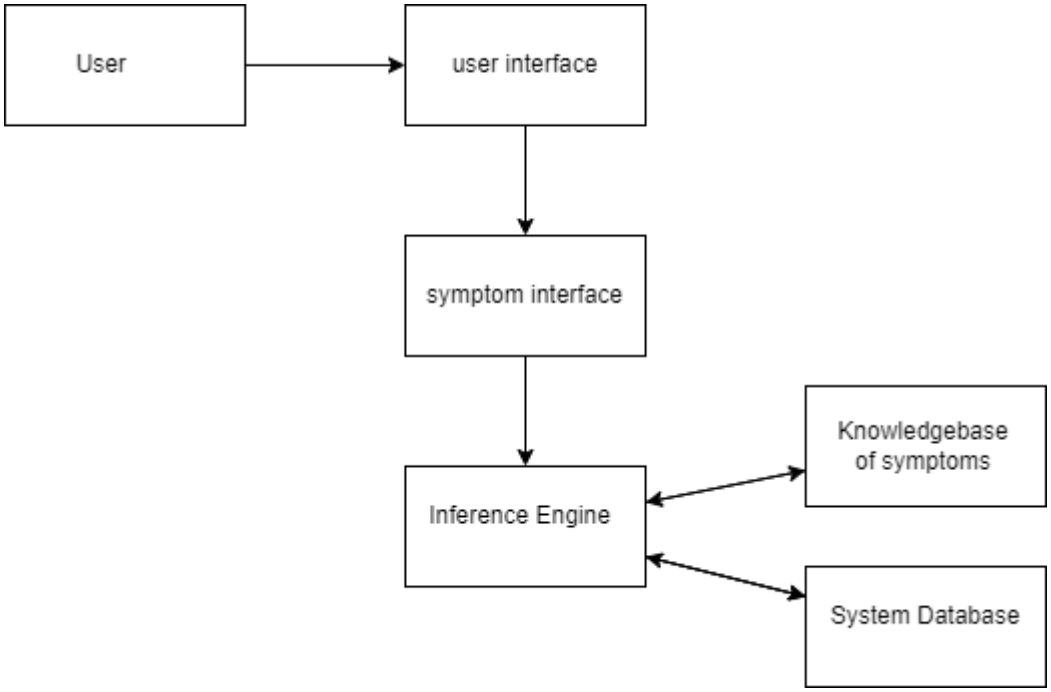
Level 1 DFD



Activity Diagram

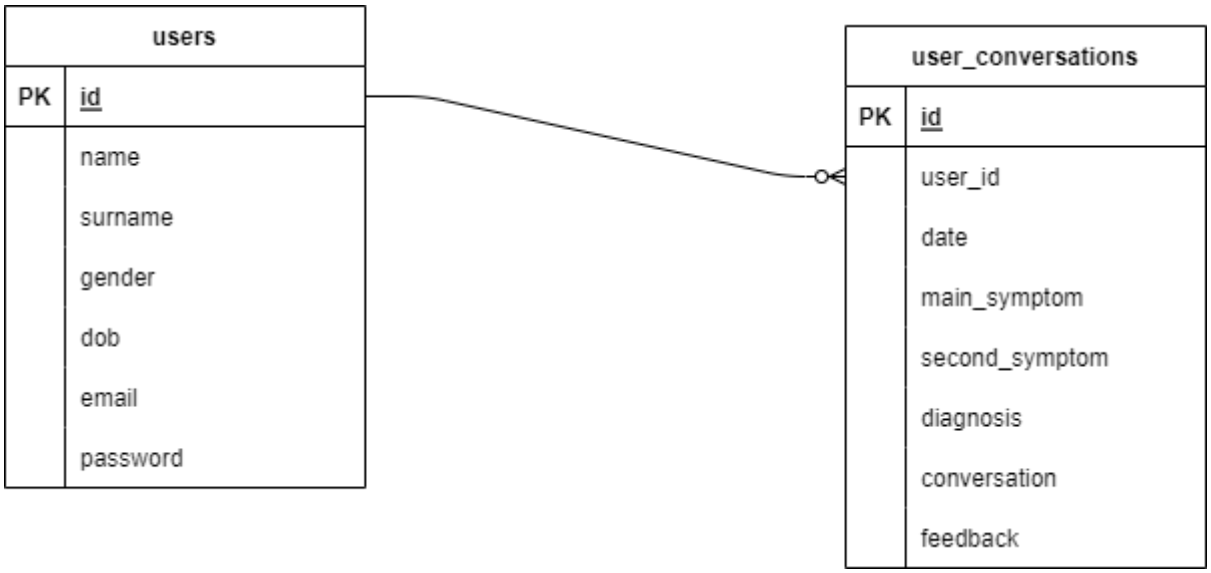

My Chiremba Activity Diagram
User
System
Registration Details
Registration Option
Store Details
Login Option
Parse Credentials
Authenticate User
Diagnosis Process
Daily Health Tips
Prompt Symptom 1
Give Description
Prompt Symptom 2
Give Description
Series of Questions
Give Answers
Perform Diagnosis
Give Results
Information on results
Store conversation

Architectural Design-hardware, networking



Database Design –ER diagrams, Normalized Databases

ER Diagram



Normalized Databases

Users Table

Field Name	Data type	Description
Name	string	The name of the user
Surname	string	The surname of the user
Gender	enum	The sex of the user, where male or female
Dob	date	The date of birth of the user, used to calculate age
Password	string	The password of the user
Email	string	The email of the user

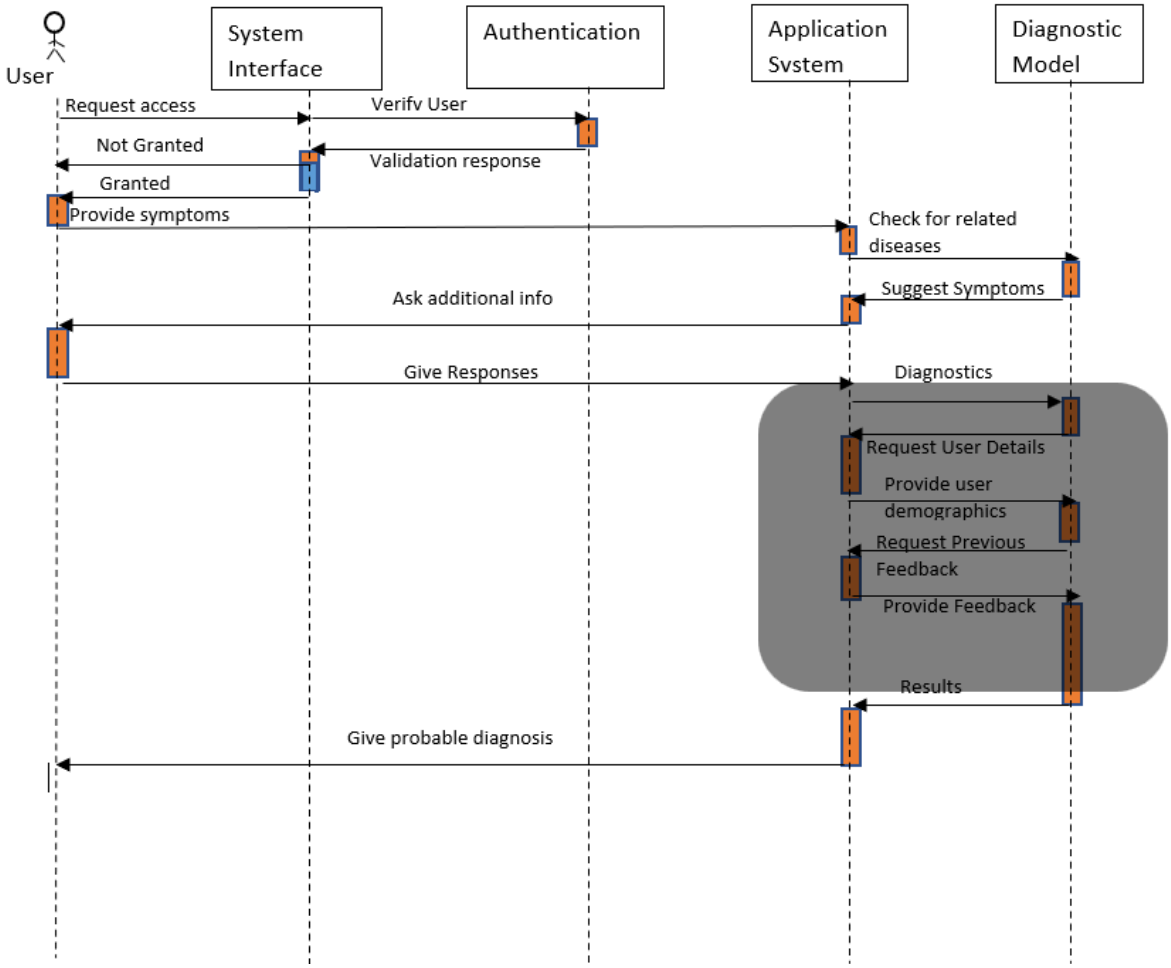
User Conversations Table

Field Name	Data type	Description
User Id	integer	The id of the user who is logged in and is having a conversation
Main Symptom	text	The main symptom that the user is feeling
Second Symptom	text	The complimentary symptom of the user, this is to help with diagnosis
Date	date	The date when the user had a conversation
Diagnosis	string	The result given by the system
Feedback	string	What the user said about his diagnosis if he said anything at all

Program Design- Sequence diagrams, Pseudo code

Sequence Diagram

Sequence Diagram of the current system

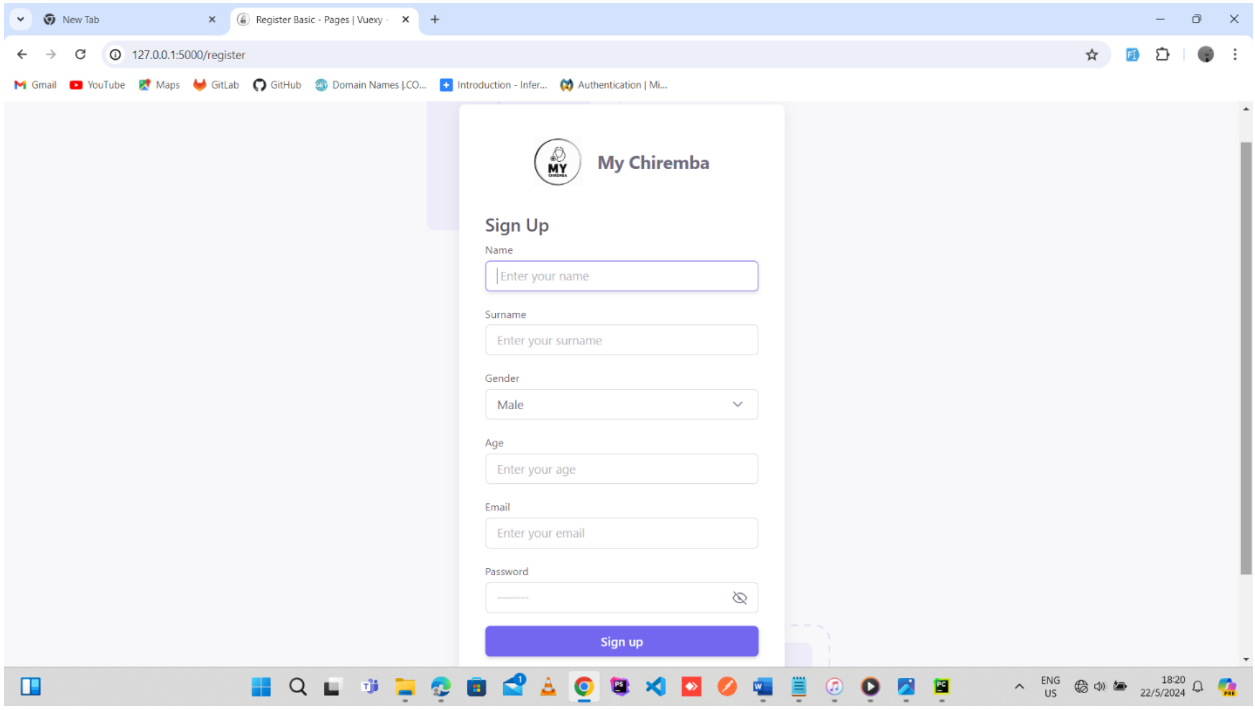


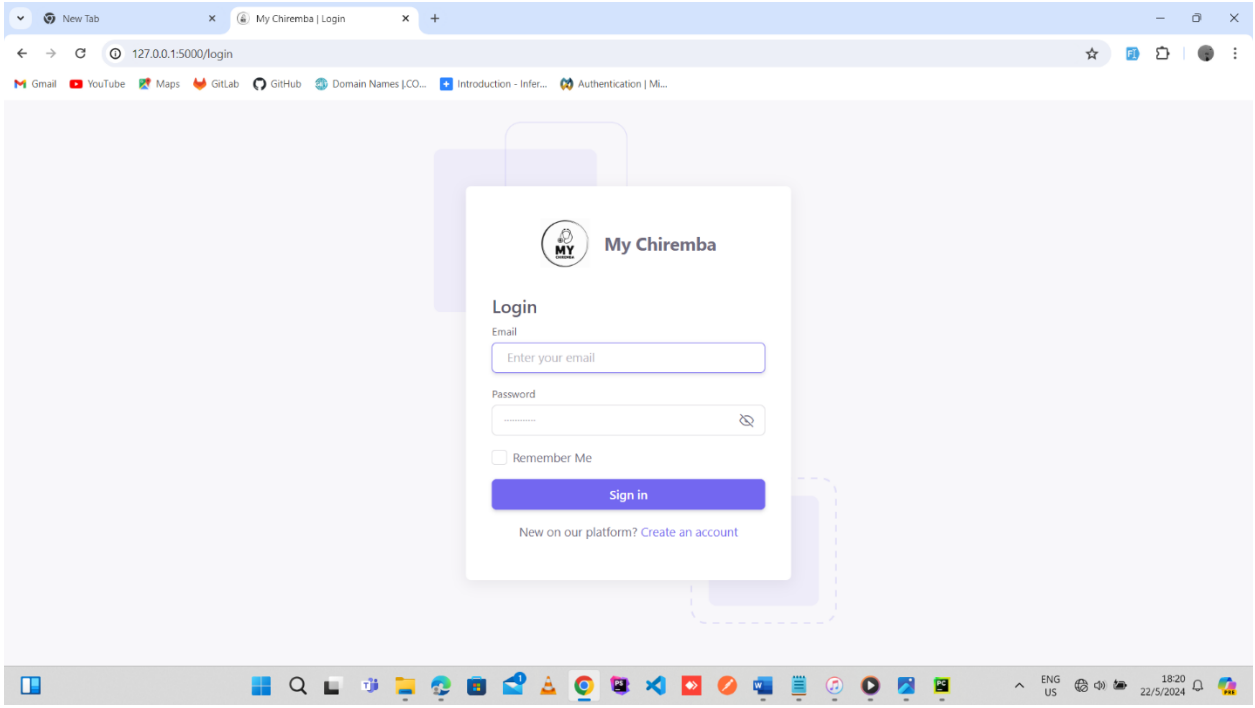
Pseudo code

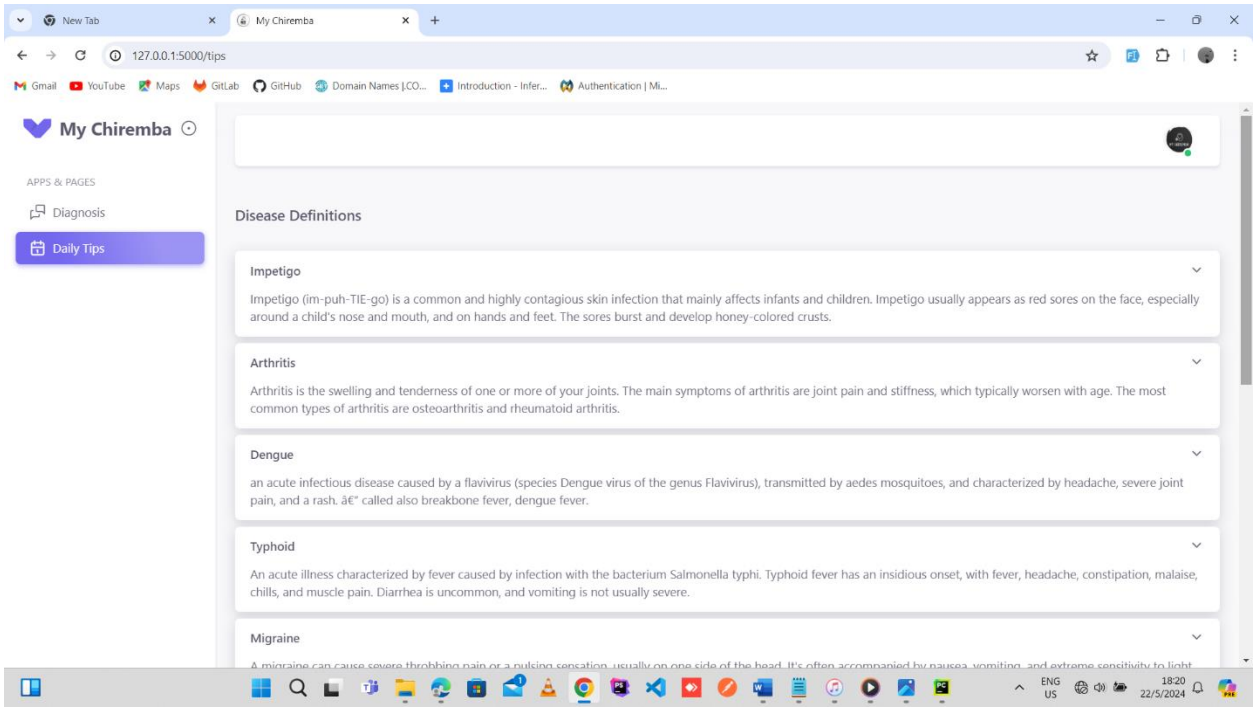
```
# Initialize the system
Initialize the AI-powered medical knowledge base
Load pre-trained machine learning models for symptom analysis, diagnosis, and
recommendation

# Main user interaction loop
while user is active:
    # Gather user input
    receive user's textual description of symptoms or health concerns
    retrieve any relevant user medical history or demographic data

    # Analyze user input
    preprocess the user's textual input
    use natural language processing to extract relevant medical concepts and
keywords
    leverage machine learning models to analyze the symptoms and generate a
list of potential diagnoses
```

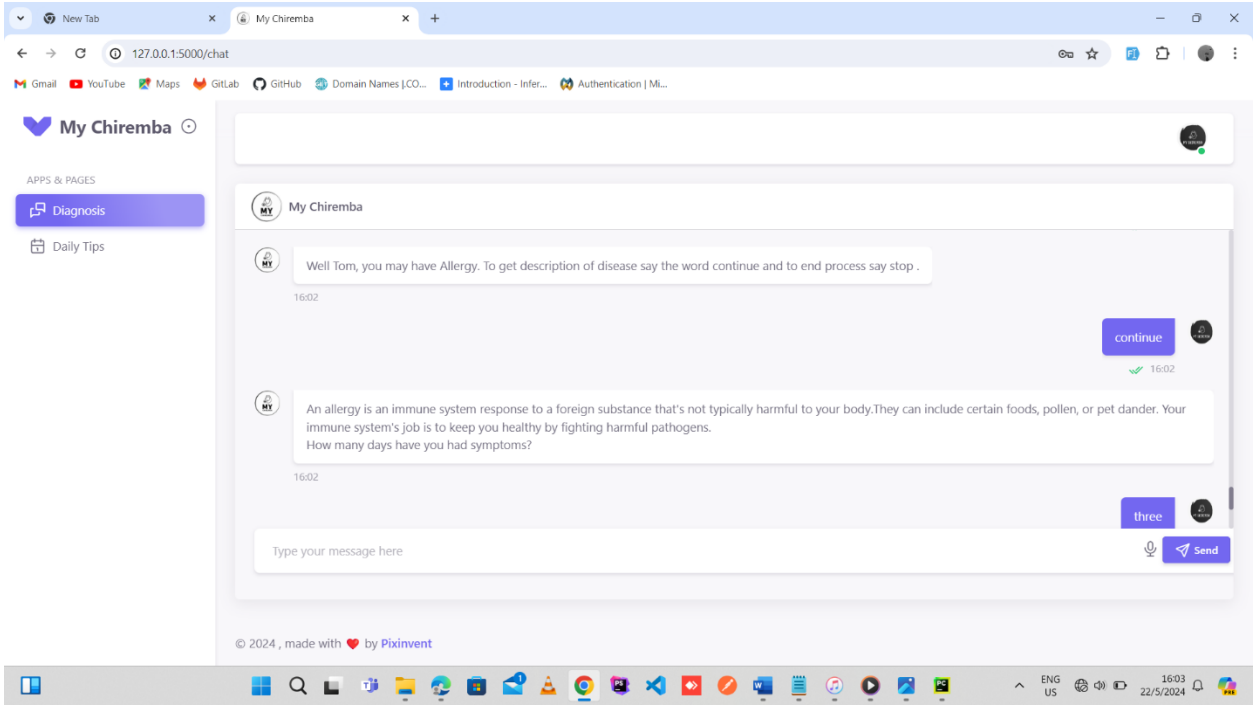
```
    # Provide personalized recommendations
    based on the potential diagnoses, retrieve relevant medical information
and treatment guidelines from the knowledge base
    use predictive analytics to recommend next steps, such as:
        - Suggested lifestyle changes or self-care actions
        - Need for further medical consultation or diagnostic testing
        - Appropriate over-the-counter remedies

    # Present results to the user
    format the recommendations in an easy-to-understand and actionable way
    display the results to the user through the my chiremba interface

    # Allow user feedback and iterative refinement
    provide the user with the ability to rate the accuracy and usefulness of
the recommendations
    use the feedback to fine-tune the machine learning models and knowledge
base over time

# System maintenance and updates
periodically update the medical knowledge base with the latest research and
clinical guidelines
retrain the machine learning models on new data to improve the system's
performance
```

Interface Design-Screenshots of user interface




New Tab
My Chiremba | Login
127.0.0.1:5000/login
Gmail
YouTube
Maps
GitLab
GitHub
Domain Names |.CO...
Introduction - Infer...
Authentication | Mi...
My Chiremba
Login
Email
Enter your email
Password
Remember Me
Sign in
New on our platform? Create an account


New Tab
My Chiremba
127.0.0.1:5000/tips
Gmail
YouTube
Maps
GitLab
GitHub
Domain Names |.CO...
Introduction - Infer...
Authentication | Mi...
My Chiremba
APPS & PAGES
Diagnosis
Daily Tips
Disease Definitions
Impetigo
Impetigo (im-puh-TIE-go) is a common and highly contagious skin infection that mainly affects infants and children. Impetigo usually appears as red sores on the face, especially around a child's nose and mouth, and on hands and feet. The sores burst and develop honey-colored crusts.
Arthritis
Arthritis is the swelling and tenderness of one or more of your joints. The main symptoms of arthritis are joint pain and stiffness, which typically worsen with age. The most common types of arthritis are osteoarthritis and rheumatoid arthritis.
Dengue
an acute infectious disease caused by a flavivirus (species Dengue virus of the genus Flavivirus), transmitted by aedes mosquitoes, and characterized by headache, severe joint pain, and a rash. â€" called also breakbone fever, dengue fever.
Typhoid
An acute illness characterized by fever caused by infection with the bacterium Salmonella typhi. Typhoid fever has an insidious onset, with fever, headache, constipation, malaise, chills, and muscle pain. Diarrhea is uncommon, and vomiting is not usually severe.
Migraine


New Tab
My Chiremba
127.0.0.1:5000/chat
Gmail
YouTube
Maps
GitLab
GitHub
Domain Names |.CO...
Introduction - Infer...
Authentication | Mi...
My Chiremba
APPS & PAGES
Diagnosis
Daily Tips
My Chiremba
Well Tom, you may have Allergy. To get description of disease say the word continue and to end process say stop .
16:02
continue
16:02
An allergy is an immune system response to a foreign substance that's not typically harmful to your body.They can include certain foods, pollen, or pet dander. Your immune system's job is to keep you healthy by fighting harmful pathogens.
How many days have you had symptoms?
16:02
three
Type your message here
Send
© 2024 , made with by Pixinvent
ENG
US
16:03
22/5/2024

CHAPTER 5: IMPLEMENTATION & TESTING

Software Testing-Unit, Module, Integration, System, Database & Acceptance

Unit Testing

1. Symptom Analyzer (Semantic Analysis)

Test Case	Input	Expected Output	Observed
1: Extract symptoms from user input	I have a headache, cough, and fever	["headache", "cough", "fever"]	Pass
2: Handle unknown symptoms	My arm hurts and I feel dizzy	["arm pain", "dizziness"]	Pass
3: Handle statement without symptom	How are you doing	[ ] *(could not extract symptom, may you please explain in other words)*	Pass

2. Diagnosis Unit

Test Case	Input	Expected Output	Observed Output
1: Generate diagnoses from known symptoms	["headache", "cough", "fever"]	["migraine", "tension headache", "sinus headache", "common cold", "flu", "bronchitis"]	Pass
2: Handle unknown symptoms	["cup", "dizziness"]	[ ] *(could not diagnose, can you please explain the symptoms better)*	Pass

3. Recommendation based on Diagnosis

Test Case	Input (Diagnosis)	Expected Output	Observed Output
1: Provide recommendations for known diagnoses	Migraine	["rest in a dark room", "stay hydrated", "get plenty of rest", "drink warm fluids"]	Pass
2: Handle uncommon diagnoses	Appendicitis	https://en.wikipedia.org/wiki/Appendicitis	Pass

Integration Testing

1. Symptom Analyzer and Diagnosis:

Test Case	Input (User Input)	Expected Output (Generated Diagnosis)	Observed
1: Handle statement with normal symptoms	I have a headache, cough, and fever	["migraine", "tension headache", "sinus headache"] You may have a migraine	Pass
2: Handle statement without symptoms	How are you doing	Hie, please describe your main symptom	Pass
3: Handle unknown symptoms	My head hurts and I am feeling dizzy	You may have hypertension	Pass
4: Handle unrelated symptoms	My teeth are bleeding and I am depressed	You are probably experiencing another symptom, please specify it	Pass

2. Symptom Analyzer, Diagnosis and Recommendations

Test Case	Input (User Input)	Expected Diagnosis	Expected Recommendation	Observed Output
1: Provide recommendations for known diagnoses	I have a headache, cough, and fever	You may have a migraine	Try to rest in a dark room, stay hydrated, get plenty of rest, drink warm fluids. Hope you get better	Pass
2: Handle uncommon diagnoses	My stomach hurts, I do not have an appetite and I have been vomiting	You may have Appendicitis	Please visit this link for more information https://en.wikipedia.org/wiki/Appendicitis	Pass

System Testing

1. End-to-End User Workflow

Test Case	Input	Expected Output	Observed Output
1. Verify the complete user workflow from symptom input to	I have a headache, muscle pain, and fatigue	1. Symptom extraction: ["headache",	Pass

diagnosis and recommendation		"muscle pain", "fatigue"] 2. Diagnosis generation: ["influenza", "mononucleosis", "viral infection"] 3. Recommendation provision: ["get plenty of rest", "stay hydrated", "take over-the-counter medication", "consider seeking medical attention"]	
2. Ensure the system handles user input with unknown symptoms and diagnoses	My leg is swollen and I have trouble breathing.	1. Symptom extraction: ["leg swelling", "difficulty breathing"] 2. Diagnosis generation: [] 3. Recommendation provision: []	Pass
3. Test the user profile management functionality	User profile creation: - Name: Jane Doe - Email: jane.doe@example.com - Age: 42	1. User profile is created successfully. 2. User profile is updated successfully.	Pass

	- Gender: Female Update user profile: - Age: 43 Delete user profile: - Email: jane.doe@example.com	3. User profile is deleted successfully.	

2. System Performance and Reliability

Test Case	Input	Expected Output	Observed Output
1: Verify the system's response time for symptom analysis and diagnosis generation	Concurrent user inputs	Response time for symptom analysis and diagnosis generation: <5 seconds	Pass
2: Ensure the system can handle a large number of user profiles and medical data	Creation of many fake user profiles with associated medical data	The system should efficiently store, retrieve, and manage the user profiles and medical data	Pass
3: Test the system's behavior under high load and stress conditions	Simultaneous user inputs	The system should remain responsive and maintain data integrity under high load	Pass

3. User Interface and Accessibility

Test Case	Input	Expected Output	Observed Output

1: Verify the user interface is intuitive and easy to navigate	User interaction with the system's UI	Users should easily understand and interact with the system's UI to input symptoms, view diagnoses, and access recommendations	Pass
2: Test the system's responsiveness and compatibility across different devices and platforms	User interaction with the system on various devices (laptop, mobile, tablet) and browsers	The system's UI and functionality are consistent and responsive across different devices and platforms	Pass

Acceptance Testing

1. Functional Acceptance Testing

Test Case	Input	Expected Output	Observed Output
1. Verify the complete user workflow from symptom input to diagnosis and recommendation	I have a headache, muscle pain, and fatigue	1. Symptom extraction: ["headache", "muscle pain", "fatigue"] 2. Diagnosis generation: ["influenza", "mononucleosis", "viral infection"]	Pass

		3. Recommendation provision: ["get plenty of rest", "stay hydrated", "take over-the-counter medication", "consider seeking medical attention"]	
2. Ensure the system handles user input with unknown symptoms and diagnoses	My leg is swollen and I have trouble breathing.	1. Symptom extraction: ["leg swelling", "difficulty breathing"] 2. Diagnosis generation: [] 3. Recommendation provision: []	Pass
3. Test the user profiles functionality	User profile creation: - Name: Jane Doe - Email: jane.doe@example.com - Age: 42 - Gender: Female Update user profile: - Age: 43 Delete user profile: - Email: jane.doe@example.com	1. User profile is created successfully. 2. User profile is updated successfully. 3. User profile is deleted successfully.	Pass

2. Non-Functional Acceptance Testing

Test Case	Input	Expected Output	Observed Output
1: Verify the system's response time for symptom analysis and diagnosis generation	Concurrent user inputs	Response time for symptom analysis and diagnosis generation: <5 seconds	Pass
2: Ensure the system can handle a large number of user profiles and medical data	Creation of many fake user profiles with associated medical data	The system should efficiently store, retrieve, and manage the user profiles and medical data	Pass
3: Test the system's behavior under high load and stress conditions	Simultaneous user inputs	The system should remain responsive and maintain data integrity under high load	Pass

3. Usability Acceptance Testing

Test Case	Input	Expected Output	Observed Output
1: Verify the user interface is intuitive and easy to navigate	User interaction with the system's UI	Users should easily understand and interact with the system's UI to input symptoms, view diagnoses, and access recommendations	Pass

2: Test the system's responsiveness and compatibility across different devices and platforms	User interaction with the system on various devices (laptop, mobile, tablet) and browsers	The system's UI and functionality are consistent and responsive across different devices and platforms	Pass

4. User Acceptance Testing

Test Case	Input	Expected Output	Observed Output
1: Conduct user acceptance testing with a representative sample of end-users	User feedback and evaluation of the system	Users should find the system to be useful, reliable, and easy to use. The system should meet their expectations and requirements.	Pass

Real code of major modules

Code for semantic analysis of the statement to get symptom from statement

```
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('stopwords')

lemmatizer = WordNetLemmatizer()

def preprocess_sent(sent):
    t = nltk.word_tokenize(sent)
    return ' '.join(
        [lemmatizer.lemmatize(w.lower()) for w in t if (w not in
set(stopwords.words('english')) and w.isalpha())])


def bag_of_words(tokenized_sentence, all_words):
    bag = np.zeros(len(all_words), dtype=np.float32)
    for idx, w in enumerate(all_words):
        if w in tokenized_sentence:
            bag[idx] = 1.0
    return bag
```

```
# predict symptom from sentence
def predictSym(sym, vocab, app_tag):
    sym = preprocess_sent(sym)
    bow = np.array(bag_of_words(sym, vocab))
    res = cosine_similarity(bow.reshape((1, -1)), df).reshape(-1)
    order = np.argsort(res)[::-1].tolist()
    possym = []
    for i in order:
        if app_tag[i].replace('_', ' ') in sym:
            return app_tag[i], 1
        if app_tag[i] not in possym and res[i] != 0:
            possym.append(app_tag[i])
    return possym, 0
```

Front end Code for Conversation

```
<form class="chatForm form-send-message d-flex justify-content-between
align-items-center" method="post">
            <div class="container-xxl flex-grow-1 container-p-y">
              <div class="app-chat card overflow-hidden">
                <div class="row g-0">
                  <!-- Chat History -->
                  <div class="col app-chat-history bg-body">
                    <div class="chat-history-wrapper">
                      <div class="chat-history-header border-bottom">
                        <div class="d-flex justify-content-between align-
items-center">
                          <div class="d-flex overflow-hidden align-items-
center">
                            <i
                              class="ti ti-menu-2 ti-sm cursor-pointer d-
lg-none d-block me-2"
                              data-bs-toggle="sidebar"
                              data-overlay
                              data-target="#app-chat-contacts"
                            ></i>
                            <div class="flex-shrink-0 avatar">
                              <img
                                src="../static/logo1.png"
                                alt="Avatar"
                                class="rounded-circle"
                                data-bs-toggle="sidebar"
                                data-overlay
                                data-target="#app-chat-sidebar-right"
                              />
                            </div>
                            <div class="chat-contact-info flex-grow-1 ms-
2">
                              <h6 class="m-0">My Chiremba</h6>
<!--                              <small class="user-status text-
muted">Symptom Helper</small>-->
                            </div>
                          </div>
                        </div>
                      </div>
                      <div class="chat-history-body bg-body">
                        <ul class="list-unstyled chat-history">
                          <li class="chat-message chat-message-right">
```

```
                            <div class="d-flex overflow-hidden">
                              <div class="chat-message-wrapper flex-grow-
1">
                                <div class="chat-message-text">
                                  <p class="mb-0">Hie</p>
                                </div>
                                <div class="text-end text-muted mt-1">
                                  <i class="ti ti-checks ti-xs me-1 text-
success"></i>
                                  <small>10:00</small>
                                </div>
                              </div>
                              <div class="user-avatar flex-shrink-0 ms-3">
                                <div class="avatar avatar-sm">
                                  <img src="../static/logo7.png"
alt="Avatar" class="rounded-circle" />
                                </div>
                              </div>
                            </div>
                          </li>
                          <li class="chat-message">
                            <div class="d-flex overflow-hidden">
                              <div class="user-avatar flex-shrink-0 me-3">
                                <div class="avatar avatar-sm">
                                  <img src="../static/logo1.png"
alt="Avatar" class="rounded-circle" />
                                </div>
                              </div>
                              <div class="chat-message-wrapper flex-grow-
1">
                                <div class="chat-message-text">
                                  <p class="mb-0">Hey my name is Chiremba.
To start diagnosis type start</p>
                                </div>
                                <div class="text-muted mt-1">
                                  <small>10:02</small>
                                </div>
                              </div>
                            </div>
                          </li>
                        </ul>
                      </div>
                      <!-- Chat message form -->
                      <div class="chat-history-footer shadow-sm input-group
input-group-merge" style="margin-right:20px">
                          <input
                            class="form-control message-input border-0 me-3
shadow-none"
                            placeholder="Type your message here"
id="chatMessage"
                          />
                          <div class="message-actions" style="margin-
right:20px">
                            <i class="speech-to-text ti ti-microphone ti-sm
cursor-pointer"></i>
                            <button type="submit" value="Submit" class="btn
btn-primary btn-sm send-msg-btn">
                              <i class="ti ti-send me-md-1 me-0"></i>
                              <span class="align-middle d-md-inline-block
d-none">Send</span>
                            </button>
```

```
                        </div>
                      </div>
                    </div>
                  </div>
                  <!-- /Chat History -->
                  <div class="app-overlay"></div>
                </div>
              </div>
            </div>
      </form>
```

Javascript Handling of Conversation

```
<script>

  const msgerForm = get(".chatForm");
  const msgerInput = get("#chatMessage");
  const msgerChat1 = get(".chat-history");

  const chatHistoryBody = document.querySelector('.chat-history-body');
  // Initialize PerfectScrollbar
  // ------------------------------


  // Chat history scrollbar
  if (chatHistoryBody) {
    new PerfectScrollbar(chatHistoryBody, {
      wheelPropagation: false,
      suppressScrollX: true
    });
  }

  // Scroll to bottom function
  function scrollToBottom() {
    chatHistoryBody.scrollTo(0, chatHistoryBody.scrollHeight);
  }
  scrollToBottom();
  msgerForm.addEventListener("submit", event => {
    event.preventDefault();

    const msgText = msgerInput.value;
    if (!msgText) return;

    appendMessage("my", msgText);
    msgerInput.value = "";
    botResponse(msgText);
  });

  function appendMessage(side, text) {
     if (side === 'my'){
    const msgHTML1 = `

                        <li class="chat-message chat-message-right">
                          <div class="d-flex overflow-hidden">
                            <div class="chat-message-wrapper flex-grow-1">
                              <div class="chat-message-text">
                                <p class="mb-0">${text}</p>
                              </div>
                              <div class="text-end text-muted mt-1">
```

```
                                <i class="ti ti-checks ti-xs me-1 text-
success"></i>
                                <small>${formatDate(new Date())}</small>
                              </div>
                            </div>
                            <div class="user-avatar flex-shrink-0 ms-3">
                              <div class="avatar avatar-sm">
                                <img src="../static/logo7.png" alt="Avatar"
class="rounded-circle" />
                              </div>
                            </div>
                          </div>
                        </li>
    `;
    msgerChat1.insertAdjacentHTML("beforeend", msgHTML1);
      scrollToBottom();
    }else{
    const msgHTML1 = `

                        <li class="chat-message">
                          <div class="d-flex overflow-hidden">
                            <div class="user-avatar flex-shrink-0 me-3">
                              <div class="avatar avatar-sm">
                                <img src="../static/logo1.png" alt="Avatar"
class="rounded-circle" />
                              </div>
                            </div>
                            <div class="chat-message-wrapper flex-grow-1">
                              <div class="chat-message-text">
                                <p class="mb-0">${text}</p>
                              </div>
                              <div class="text-muted mt-1">
                                <small>${formatDate(new Date())}</small>
                              </div>
                            </div>
                          </div>
                        </li>
    `;
    msgerChat1.insertAdjacentHTML("beforeend", msgHTML1);
    msgerChat1.scrollTop += 500;

      scrollToBottom();
    }
  }

  function botResponse(rawText) {
    // Bot Response
    $.get("/get", { msg: rawText }).done(function (data) {
      console.log(rawText);
      console.log(data);
      const msgText = data;
      appendMessage("friend", msgText);

    });

  }


  // Utils
  function get(selector, root = document) {
    return root.querySelector(selector);
```

```
  }

  function formatDate(date) {
    const h = "0" + date.getHours();
    const m = "0" + date.getMinutes();

    return `${h.slice(-2)}:${m.slice(-2)}`;
  }



</script>
```

Code For the whole conversation session (asking question, getting diagnosis, description and precaution)

```
@app.route("/get")
def get_bot_response():
    try:
        s = request.args.get('msg')
        if session['diagnosis'] == False and s.upper() != "START" and
s.upper() != "RESTART":
            return "Hie to begin the process type start"
        if s.upper() == "START":
            # session['name'] = 'Simon'
            session['step'] = "FS"  # first symp

            session['diagnosis'] = True
            return "Well, Hello  " + session[
                "name"] + ", I will be asking some few questions about your
symptoms to see what you should do. Can you describe your main symptom  " +
\
                session["name"] + " ?"
        if s.upper() == "RESTART":
            name = session['name']
            session.clear()
            session['name'] = name
            session['diagnosis'] = False
            return "You have successfully restarted session"
        if session['step'] == "FS":
            sym1 = s
            psym1, find = predictSym(sym1, vocab, app_tag)
            if find == 1:
                sym1 = psym1
                session['step'] = 'SS'
                message = "You are probably facing another symptom, if so,
can you specify it?"
            else:
                try:
                    message = 'Do you experience ' + psym1[0].replace('_',
' ')
                    session['step'] = 'FSClarification'
                    session['FSClarificationLimit'] = len(psym1)
                    session['FSClarificationCount'] = 1
                except:
                    return "Could not find your stated symptom in our
dataset. please explain the symptom in simpler terms"
            session['FST'] = [sym1, psym1]
            return message
        if session['step'] == "FSClarification":
```

```
            temp = session['FST']
            if s != 'yes' and s != 'no':
                return 'Please enter only the values yes and no'
            elif s == 'yes':
                sym1 = temp[1][session['FSClarificationCount'] - 1]
                session['FST'] = [sym1, temp[1]]
                session['step'] = 'SS'
                message = "You are probably facing another symptom, if so,
can you specify it?"
            else:
                if session['FSClarificationCount'] >
session['FSClarificationLimit']:
                    session['step'] = 'BYE'
                    message = "Sorry could not verify symptom, please try
again later"
                else:
                    message = 'Do you experience ' +
temp[1][session['FSClarificationCount']].replace('_', ' ')
                    session['FSClarificationCount'] += 1
            return message
        if session['step'] == "SS":
            sym2 = s
            psym2, find = predictSym(sym2, vocab, app_tag)
            if find == 1:
                sym2 = psym2
                session['AS'] = [session['FST'][0], sym2]
                session['diseases'] = possible_diseases(session['AS'])
                session['diseasesno'] = len(session['diseases'])
                session['diseasescount'] = 0
                session['step'] = 'SDF'
                message = "Okay, from now on you are supposed to provide
only the answers yes and no. Say yes if you understand"
            else:

                try:
                    message = 'Do you experience ' + psym2[0].replace('_',
' ')
                    session['step'] = 'SSClarification'
                    session['SSClarificationLimit'] = len(psym2)
                    session['SSClarificationCount'] = 1
                except:
                    return "Could not find your stated symptom in our
dataset. please explain the symptom in simpler terms"
            session['SST'] = [sym2, psym2]
            return message
        if session['step'] == "SSClarification":
            temp = session['SST']
            if s != 'yes' and s != 'no':
                return 'Please enter only the values yes and no'
            elif s == 'yes':
                sym2 = temp[1][session['SSClarificationCount'] - 1]
                session['SST'] = [sym2, temp[1]]
                session['AS'] = [session['FST'][0], sym2]
                session['diseases'] = possible_diseases(session['AS'])
                session['diseasesno'] = len(session['diseases'])
                session['diseasescount'] = 0
                session['step'] = 'SDF'
                message = "Okay, from now on you are supposed to provide
only the answers yes and no. Say yes if you understand"
            else:
                if session['SSClarificationCount'] >
```

```
session['SSClarificationLimit']:
                    session['step'] = 'BYE'
                    message = "Sorry could not verify symptom, please try
again later"
                else:
                    message = 'Do you experience ' +
temp[1][session['SSClarificationCount']].replace('_', ' ')
                    session['SSClarificationCount'] += 1
            return message
        if session['step'] == "SDF":
            if s != 'yes' and s != 'no':
                return 'Please enter only the values yes and no'
            elif s == 'yes':
                session['step'] = 'SD'
            else:
                session['step'] = 'BYE'
                message = 'You did not accept to continue therefore the
session has closed'
                return message
        if session['step'] == "SD":
            session['SDStop'] = False
            disease = session['diseases'][session['diseasescount']]
            session['symdisease'] = []
            for sym in symVONdisease(df_tr, disease):
                if sym not in session['AS']:
                    session['symdisease'].append(sym)
            session['symdiseaselim'] = len(session['symdisease'])
            session['symdiseasescount'] = 1
            session['step'] = 'SDM'
            return "Are you experiencing any " + session['symdisease'][0]
        if session['step'] == "SDM":
            if s != 'yes' and s != 'no':
                return 'Please enter only the values yes and no'
            elif s == 'yes':

session['AS'].append(session['symdisease'][session['symdiseasescount'] -
1])
                dise = possible_diseases(session['AS'])
                if len(dise) == 1:
                    session['SDStop'] = True
                if session['SDStop'] or session['diseasescount'] >
session['diseasesno']:
                    session['step'] = 'Predict'
                elif session['symdiseasescount'] >
session['symdiseaselim']:
                    session['step'] = 'SD'
                    session['diseasescount'] += 1
                else:
                    session['symdiseasescount'] += 1
                    print(session['symdiseasescount'])
                    return "Are you experiencing any " +
session['symdisease'][session['symdiseasescount'] - 1]
            else:
                session['symdiseasescount'] += 1
                return "Are you experiencing any " +
session['symdisease'][session['symdiseasescount'] - 1]
        if session['step'] == "Predict":
            result = knn.predict(OHV(session['AS'], all_symp_col))
            if result is None:
                session['step'] = 'BYE'
                return 'Sorry, Could not find prediction. Please try again
```

```
later'
            session['result'] = result[0]
            session['step'] = 'Description'
            return "Well  " + session["name"] + ", you may have " +
session[
                'result'] + ". To get description of disease say the word
continue and to end process say stop ."
        if session['step'] == "Description":
            session['step'] = "Severity"
            if session['result'] in description_list.keys():
                return description_list[session['result']] + " \n <br>  How
many days have you had symptoms?"
            else:
                if " " in session['result']:
                    session['result'] = session['result'].replace(" ", "_")
                    session['step'] = 'FINAL'
                return "please visit https://en.wikipedia.org/wiki/" +
session['result'] + "  for more information"
        if session['step'] == "Severity":
            session['step'] = 'FINAL'
            if calc_condition(session['AS'], int(convert_word_to_int(s)))
== 1:
                return "you should take the consultation from doctor <br>
say quit to exit"
            else:
                msg = 'Nothing to worry about, but you should take the
following precautions :<br> '
                i = 1
                for e in precautionDictionary[session["result"]]:
                    msg += '\n ' + str(i) + ' - ' + e + '<br>'
                    i += 1
                msg += ' say quit to end'
                return msg
        if session['step'] == "FINAL":
            session['step'] = "BYE"
            return "Your diagnosis was perfectly completed. Do you need
another medical consultation (yes or no)? "
        if session['step'] == "BYE":
            name = session["name"]
            session.clear()
            session['name'] = name
            session['diagnosis'] = False
            if s.lower() == "yes":
                session["name"] = name
                session['step'] = "FS"
                return "HELLO again  " + session["name"] + " Please tell me
your main symptom. "
            else:
                return "Thank you  " + name + ". Enjoy your day"
    except Exception as e:
        print(e)
        return "Sorry, can not get idea of your symptom"
```

Functions For Diagnosis description, precaution and severity of condition

```
def getDescription():
    global description_list
    with open('Medical_dataset/symptom_Description.csv') as csv_file:
        csv_reader = csv.reader(csv_file, delimiter=',')
        for row in csv_reader:
```

```
            _description = {row[0]: row[1]}
            description_list.update(_description)


def getSeverityDict():
    global severityDictionary
    with open('Medical_dataset/symptom_severity.csv') as csv_file:

        csv_reader = csv.reader(csv_file, delimiter=',')
        try:
            for row in csv_reader:
                _diction = {row[0]: int(row[1])}
                severityDictionary.update(_diction)
        except:
            pass


def getprecautionDict():
    global precautionDictionary
    with open('Medical_dataset/symptom_precaution.csv') as csv_file:
        csv_reader = csv.reader(csv_file, delimiter=',')
        for row in csv_reader:
            _prec = {row[0]: [row[1], row[2], row[3], row[4]]}
            precautionDictionary.update(_prec)
```

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

Results and summary, Recommendations & Future Works

Results

The system provided proper diagnosis and further provided complementary information to better understand and address the disease being faced

Objectives	Fully achieved	Partially achieved
To provide real-time interaction which uses natural language processing for semantic analysis of the description of symptoms	✓	
To provide a functionality for diagnosing a user based on given symptom	✓	
To provide a functionality for giving the users daily health advices	✓	

Summary

The My Chiremba system is an innovative healthcare assistant application that aims to revolutionize the way individuals manage their personal health and wellness. By leveraging advanced natural language processing and machine learning algorithms, the system empowers users to gain personalized insights into their symptoms, potential diagnoses, and tailored recommendations. With a user-friendly interface and a comprehensive knowledge base, My Chiremba seamlessly guides users through the process of understanding their health concerns, facilitating informed decision-making and early intervention, where appropriate. The system's robust architecture, data management capabilities, and focus on accessibility and usability have been meticulously designed to provide a transformative experience for individuals seeking to take a more proactive approach to their healthcare.

Recommendations and Future Work

1. Medical Image Analysis: The system will be able to diagnose a user based on the picture of the skin disease he would have uploaded on the system
2. Multilingual assistance: The system should allow communication even using languages like Shona and Ndebele
3. Nearest Doctor's Assistance: When the system would have determined the criticality of the disease, it should be able to recommend a specific doctor well vest with that particular disease and the doctor should have an account on the system

Conclusion

As the healthcare landscape continues to evolve, My Chiremba stands poised to play a pivotal role in empowering patients, reducing unnecessary medical costs, and fostering a more engaged and informed population. By democratizing access to medical expertise and putting the power of personalized healthcare guidance in the hands of users, the project seeks to redefine the way individuals manage their wellbeing and interact with the healthcare system.

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Appendices

Sample Of Data Collection

Sample 1: Tinashe Chaitezvi

Questions Responses 3 Settings

My Chiremba

The questions below are going to be used to access the current medical treatment process and the evaluate the need for a new automated system which helps with that so may you please be sincere in answering the question. Please answer all of the questions so that we may have a better reflection, thank you.

Enter your name

Tinashe Chaitezvi

1. How do you typically schedule appointments for medical diagnosis?

- (●) Phone call
- ( ) In-person visit
- ( ) Online portal
- ( ) Other:

2. What challenges do you face during the appointment scheduling process? (Select all that apply)

- ( ) Difficulty reaching the healthcare provider
- ( ) Long waiting times to get an appointment
- ( ) Limited availability of preferred healthcare providers
- (●) Other: Money required handina

3. On average, how long do you typically wait for your appointment?

- ( ) Less than 5 minutes
- ( ) 5-30 minutes
- ( ) 30 minutes to 1 hour
- (●) More than 1 hour

Questions Responses 3 Settings

4. How would you rate the overall efficiency of the current procedure for medical diagnosis?

- ○ Very efficient
- ○ Somewhat efficient
- ○ Neutral
- ◉ Inefficient

5. Would you support the introduction of a medical system that assists with the medical diagnosis and treatment process to compliment the manual system?

- ◉ Yes
- ○ No

6. What do you like about the manual system?

Nothing

7. What do you dislike about the current manual system?

Everything

8. Is there anything else you would like to share regarding your experience with the current system?

SLow

9. What improvements or changes would you suggest to enhance the current procedure for medical diagnosis?

Supply equipment

Sample 2: Anonymous

Questions Responses 3 Settings

Responses cannot be edited

My Chiremba

The questions below are going to be used to access the current medical treatment process and the evaluate the need for a new automated system which helps with that so may you please be sincere in answering the question. Please answer all of the questions so that we may have a better reflection, thank you.

Enter your name

.....

1. How do you typically schedule appointments for medical diagnosis?

- ○ Phone call
- ◉ In-person visit
- ○ Online portal
- ○ Other: .....

2. What challenges do you face during the appointment scheduling process? (Select all that apply)

- ○ Difficulty reaching the healthcare provider
- ○ Long waiting times to get an appointment
- ◉ Limited availability of preferred healthcare providers
- ○ Other: .....

3. On average, how long do you typically wait for your appointment?

- ○ Less than 5 minutes
- ○ 5-30 minutes
- ○ 30 minutes to 1 hour
- ◉ More than 1 hour

Questions Responses 3 Settings

4. How would you rate the overall efficiency of the current procedure for medical diagnosis?

- (●) Very efficient
- ( ) Somewhat efficient
- ( ) Neutral
- ( ) Inefficient

5. Would you support the introduction of a medical system that assists with the medical diagnosis and treatment process to compliment the manual system?

- (●) Yes
- ( ) No

6. What do you like about the manual system?

its personal

7. What do you dislike about the current manual system?

it takes too long

8. Is there anything else you would like to share regarding your experience with the current system?

n/a

9. What improvements or changes would you suggest to enhance the current procedure for medical diagnosis?

make it personal amd fast

Sample 3: Tariro Moyo

Questions Responses 3 Settings

Responses cannot be edited

My Chiremba

The questions below are going to be used to access the current medical treatment process and the evaluate the need for a new automated system which helps with that so may you please be sincere in answering the question. Please answer all of the questions so that we may have a better reflection, thank you.

Enter your name

Tariro Moyo

1. How do you typically schedule appointments for medical diagnosis?

- ○ Phone call
- ◉ In-person visit
- ○ Online portal
- ○ Other:

2. What challenges do you face during the appointment scheduling process? (Select all that apply)

- ○ Difficulty reaching the healthcare provider
- ○ Long waiting times to get an appointment
- ◉ Limited availability of preferred healthcare providers
- ○ Other:

3. On average, how long do you typically wait for your appointment?

- ○ Less than 5 minutes
- ○ 5-30 minutes
- ○ 30 minutes to 1 hour
- ◉ More than 1 hour

Questions Responses 3 Settings

4. How would you rate the overall efficiency of the current procedure for medical diagnosis?

- ○ Very efficient
- ◉ Somewhat efficient
- ○ Neutral
- ○ Inefficient

5. Would you support the introduction of a medical system that assists with the medical diagnosis and treatment process to compliment the manual system?

- ◉ Yes
- ○ No

6. What do you like about the manual system?

The doctors and nurses are nice, they take time to understand your situation without being impatient

7. What do you dislike about the current manual system?

there are very long queues

8. Is there anything else you would like to share regarding your experience with the current system?

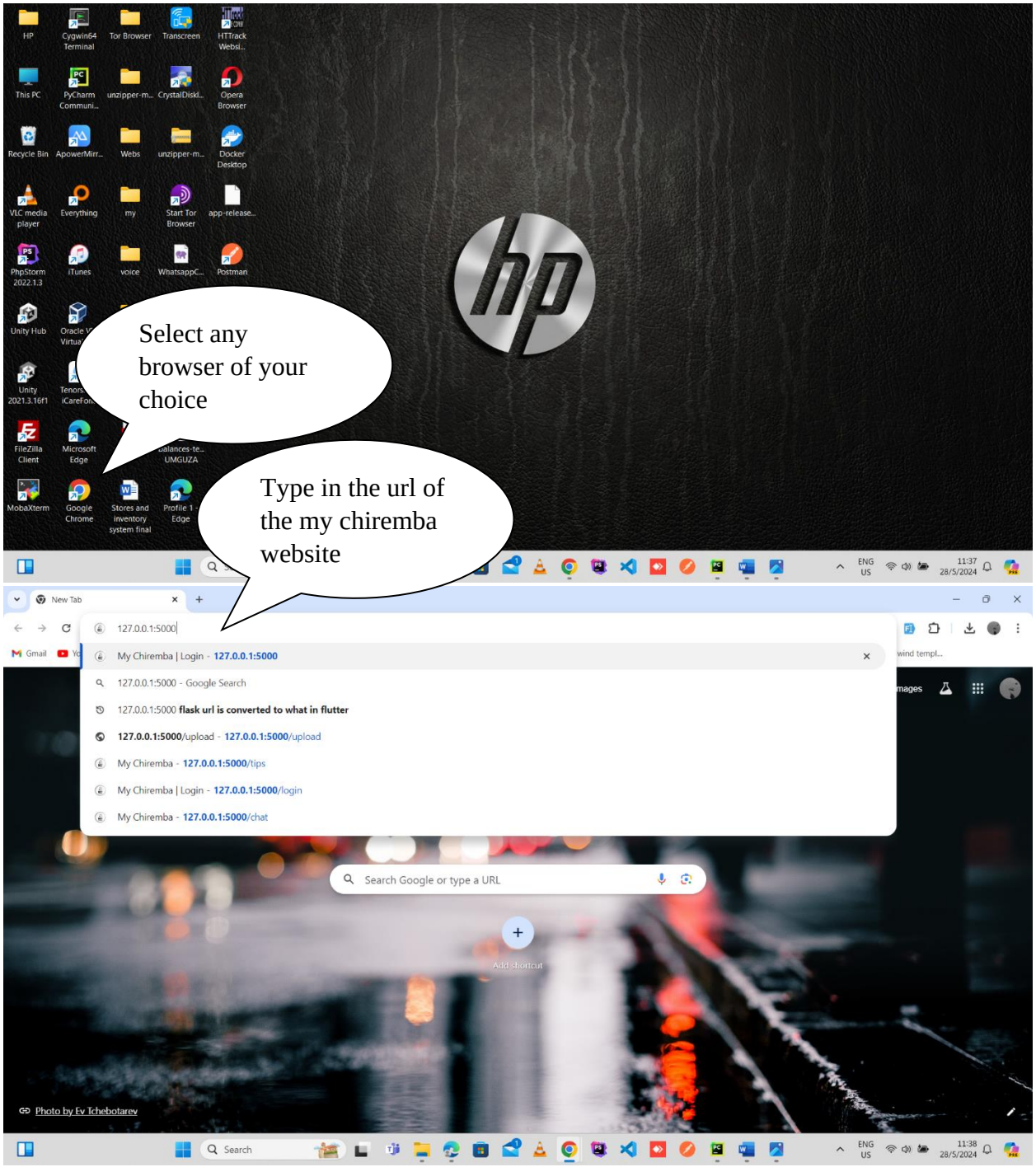
i rarely go for checkups because of money

9. What improvements or changes would you suggest to enhance the current procedure for medical diagnosis?

to improve on time

User Manual

Navigation



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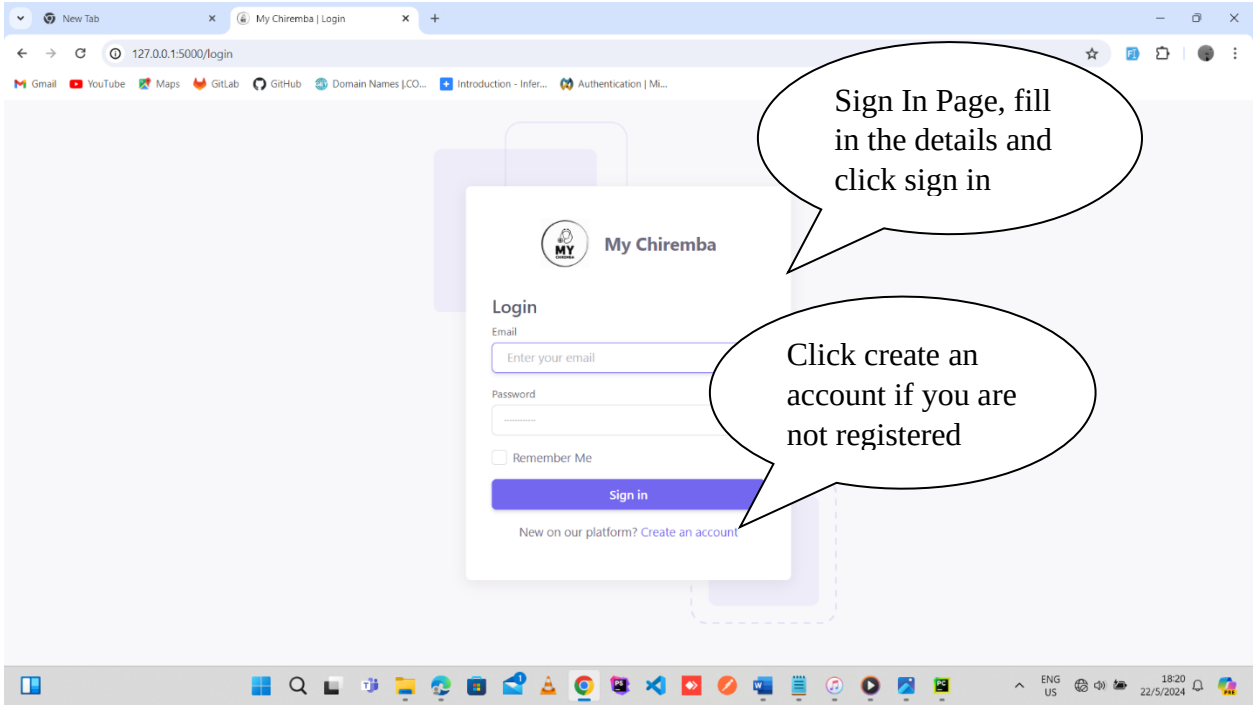
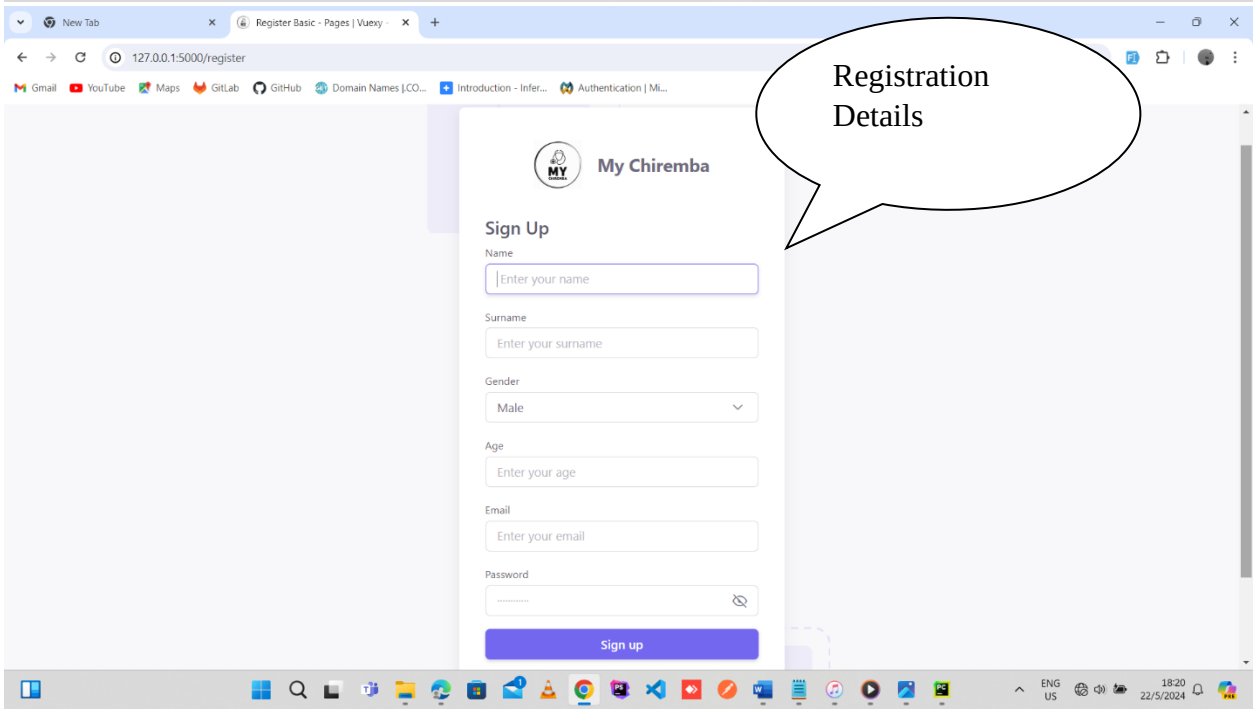
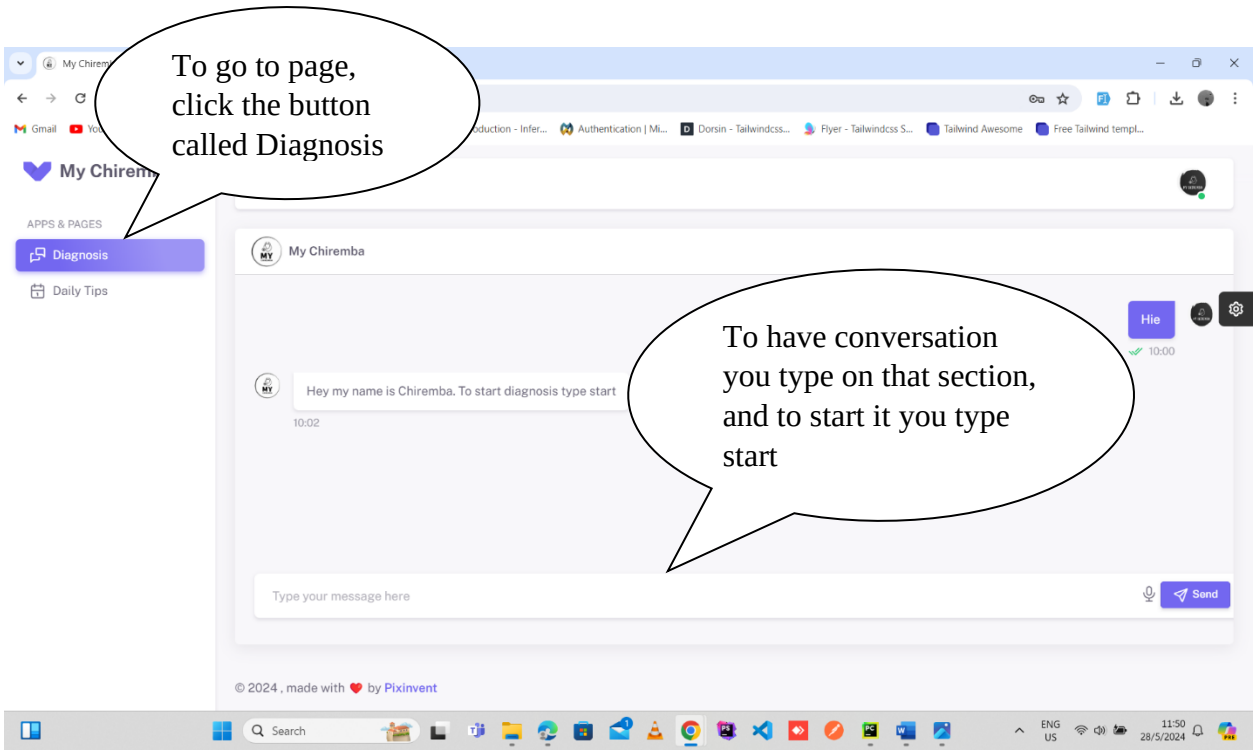
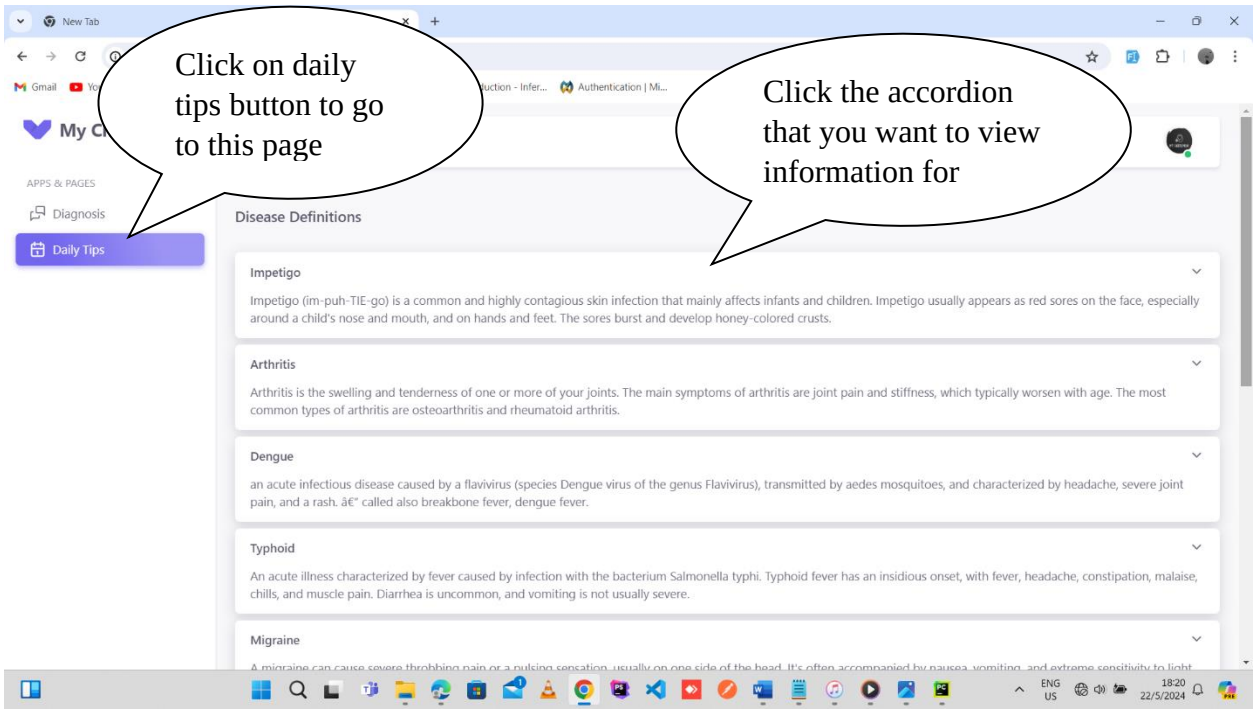




System Diagnosis Page



Daily Tips



Technical Paper

MY CHIREMBA: APP FOR MEDICAL DIAGNOSIS

Bachelor of Technology Honors Degree in Software Engineering

of the

School Of Information Sciences and Technology

at

Harare Institute of Technology

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Abstract

This document proposes the next generation technology that everyone has been waiting for, a novel system that provides medical diagnosis assistance to users. The system is called My Chiremba. The system is designed to help users understand their symptoms and possible causes by guiding them through a series of questions to help narrow down symptoms to a possible diagnosis. The document also entails the root problems that drove the innovation of the system, investigations undergone and the objectives and scope that My Chiremba will have to meet.

Introduction

The rapid pace of technological innovation has transformed nearly every aspect of our lives, revolutionizing the way we work, communicate, and even manage our health and wellbeing. In the field of healthcare, the integration of cutting-edge tools and techniques has ushered in a new era; of more efficient, effective, and personalized care; that promises to deliver more accurate diagnoses, targeted treatments, and preventive care tailored to the unique needs of each individual.

However, despite these strides, a stark reality persists. In many communities around the world, including Zimbabwe, a significant portion of the population continues to face daunting barriers when it comes to accessing quality healthcare services. Whether it's the prohibitive cost of visiting a doctor or the lack of essential medical equipment and diagnostic tools in underfunded healthcare facilities, these challenges have left far too many Zimbabweans with limited options for managing their health.

Evidence of the above claim is given by the research information below.

1. In an Al Jezeera article written on 28 April 2021 with the title *Rich or poor, in Zimbabwe crumbling healthcare is deadly for all*, the researcher tells the story of a Chitungwiza resident called Leonard Mugwade, who accompanied his neighbour to Chitungwiza Central Hospital where they were turned away because the neighbour was said to have had a non-crucial case, but then died before the Leonard could get $25 to take the person to another private hospital.

2. In a Herald article written on 03 January 2020 with the title *public hospitals adjust charges*, research showed that as of that time the consultation fees for adults receiving treatment at Parirenyatwa Group of Hospitals are up from $15 to $200 for initial and $130 for subsequent visits. Children above five years would pay $100 as initial consultation and $60 for reviews. In light of this information Mr Rusike then concluded by saying, *the low-income population can no longer afford to get ill because they cannot afford expensive curative services*.

This unacceptable state of affairs underscores the urgent need for innovative solutions that address the systemic barriers to quality healthcare, ensuring that every individual, regardless of their financial situation or geographical location, can access the care they deserve.

Problem Statement

Many patients lack access to medical professionals who provide accurate and timely diagnosis and treatment. This can be due to financial difficulties or the patients staying in remote or underserved areas where there are no nearby clinics or hospitals. (**It is costly**)

Healthcare providers often face time constraints in conducting comprehensive assessments and consultations with patients. The limited time available can lead to delays in treatment and ultimately poorer health outcomes for clients. (**It takes too much time**)

Related Work

An article was published in the National Library of medicine website with the title *Chatbots and Conversational Agents in Mental Health: A review of the Psychiatric Landscape*. The results of the study showed that conversational agents had potential for benefit in psychoeducation and self-adhering and also satisfaction rating of chatbots were extremely high, therefore suggesting that it would be an effective and at the same time enjoyable tool in psychiatric treatment

Artificial Intelligence-Based Symptom Checker Systems: A Review and Critical Appraisal by Johnson et al. (2020): This review article explores the existing AI-based symptom checker systems and evaluates their accuracy, usability, and clinical relevance. It discusses the challenges associated with symptom analysis and provides insights into the potential of AI-driven

systems for improving diagnosis and decision-making in healthcare.

Another article on with the title *Mobile Health Applications for Diagnosis and Management of Medical Conditions* by Lalloo et al. (2019) examines the use of mobile health (mHealth) applications for diagnosis and management of medical conditions. It discusses the role of mHealth apps in symptom tracking, self-diagnosis, and remote monitoring of patients. The review also addresses the challenges and opportunities associated with mHealth app adoption in healthcare.

Viz.ai is a mobile app that uses artificial intelligence to detect diseases such as intracerebral haemorrhage and vessel occlusions. This app also provides mobile imaging and HIPPA-complaint messaging to help synchronize workflow across many specialties for diseases such as stroke.

EpicCare is an electronic record system that allows health professionals to update patient's records in real time and also allows for the integration of various medical devices and data sources.

Infermedica offers an AI-driven symptom checker that enables users to describe their symptoms. The system analyzes the input and provides potential diagnoses and recommendations for further medical evaluation

Buoy Health is a web based ai health assistant that provides personalized clinical support and helps people make the best decisions about their health. It gives people insight on what to do when they are sick

Solution

It is against this background that the My Chiremba project emerges as a beacon of hope, giving a digital health solution that leverages the power of artificial intelligence to bring the benefits of personalized healthcare guidance directly to the fingertips of users. The objectives are:

- To provide real-time interaction which uses natural language processing for semantic analysis of the description of symptoms
- To provide a functionality for diagnosing a user based on given symptom
- To provide a functionality for giving the users daily health advices

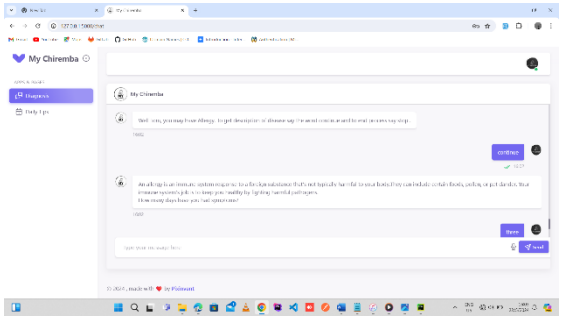
And the benefits are:

- Improved Healthcare Accessibility: Many people in Zimbabwe do not have access to healthcare facilities or cannot afford medical care. By leveraging the proposed technology, the My Chiremba can help bridge this gap by providing healthcare assistance to individuals who may have limited access to medical professionals or healthcare facilities, particularly in remote or underserved areas.

- Early Disease Detection: The timely identification and diagnosis of diseases significantly impact patient outcomes. My Chiremba can serve as an efficient screening tool, enabling individuals to receive preliminary assessments and recommendations promptly, leading to early detection and intervention. With this the app can also help reduce the burden on healthcare facilities by providing pre-primary care for quick guidance so that red flags are timely detected by doctors and patients

- Personalized Healthcare: The app can utilize artificial intelligence algorithms to analyze individual symptoms, medical history, and other relevant factors to provide personalized insights and recommendations to users, empowering them to make informed decisions and take proactive measures for their health.

Results and Future Works



The system provided proper diagnosis and further provided complementary information to better understand and address the disease being faced

Objectives	Fully achieved	Partially achieved
To provide real-time interaction which uses natural language processing for semantic analysis of the description of symptoms	✓	
To provide a functionality for diagnosing a user based on given symptom	✓	
To provide a functionality for giving the users daily health advices	✓	

Future Works

4. Medical Image Analysis: The system will be able to diagnose a user based on the picture of the skin disease he would have uploaded on the system
5. Multilingual assistance: The system should allow communication even using languages like Shona and Ndebele
6. Nearest Doctor's Assistance: When the system would have determined the criticality of the disease, it should be able to recommend a specific doctor well vest with that particular disease and the doctor should have an account on the system

Conclusion

As the healthcare landscape continues to evolve, My Chiremba stands poised to play a pivotal role in empowering patients, reducing unnecessary medical costs, and fostering a more engaged and informed population. By democratizing access to medical expertise and putting the power of personalized healthcare guidance in the hands of users, the project seeks to redefine the way individuals manage their wellbeing and interact with the healthcare system.

Acknowledgement

First and foremost, I would like to thank the Lord Almighty for giving me the strength and wisdom to be able to complete my capstone project successfully.

I am forever indebted to my parents for showing their unwavering support; be it financial, physical, emotional and spiritual support; during this course of learning curvature.

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The project's success is also a testament to the contributions of the various collaborators, stakeholders and other participants, who helped in some idea generation, data collection and support that they provided.

It is through the collective efforts of all the participants above that drove to the successful completion of the My Chiremba 2024 capstone project.

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THANK YOU